

Software-as-a-Graph: Heterogeneous Graph Learning for Pre-Deployment Dependability Analysis of Complex Distributed Systems

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Abstract

Modern asynchronous, publish-subscribe, and microservice architectures pose a critical pre-deployment visibility barrier: bug-free service code can still harbor catastrophic outages from hidden single points of failure or mismatched middleware contracts, a disconnect termed the Architecture-Code Gap. To evaluate reliability before deployment without runtime telemetry, we present Software-as-a-Graph (SaG), a static framework converting Architecture-as-Code manifests into typed multigraphs across five core entity types. SaG operates through two decoupled pathways: a relation-specific Heterogeneous Graph Transformer with Quality-of-Service edge encodings (HGT-QoS) forecasting cascade blast radii, and an interpretable ISO/IEC 25010 attribution layer guiding repairs.

Evaluated across twelve inductive scenarios and five open-source systems against simulation oracles, relation typing improves Spearman rank correlation by $\Delta\rho = +0.114$ over homogeneous learning under distribution shift ($p = 0.0122$, winning 11 of 12 folds), whereas in-distribution performance is indistinguishable. QoS edge encodings contribute an independent $+0.054$ ($p = 0.0093$). Against training-free QoS-weighted centrality, the learned model does not establish a statistically significant ranking advantage ($+0.127$, $p = 0.077$), indicating that teams requiring only scalar rankings can rely on topological heuristics. Zero-shot transfer achieves $\rho = 0.680$ overall, but drops to $+0.160$ on active failure-propagating components. Forward inference takes under 56 ms, leaving deterministic structural extraction as the primary bottleneck. Ultimately, training a deep heterogeneous model is justified not by scalar ranking alone, but by its relational inductive bias under distribution shift, per-relationship edge criticalities, and attention maps across heterogeneous schemas.

Keywords: Heterogeneous graph neural networks, Distributed systems dependability, Publish-subscribe architecture, Cascading failures, Static system analysis, Explainable AI

1. Introduction

1.1. Motivation

Modern large-scale distributed software systems increasingly rely on asynchronous, event-driven, and publish-subscribe (pub-sub) architectures. Across diverse domains—from autonomous driving (ROS 2 [1]) and enterprise event streams (Apache Kafka [2]) to cyber-physical backbones (DDS [3]), IoT fleets (MQTT [4]), cloud-native microservices [5, 6], and distributed AI/LLM serving clusters—pub-sub decouples producers and consumers in space, time, and synchronization [7]. Components interact indirectly through intermediate message topics and brokers without maintaining direct static references. Furthermore, modern middleware specifications allow engineers to configure deployment-time Quality-of-Service (QoS) policies—such as reliability guarantees, durability, message priorities, and delivery deadlines—to govern how traffic behaves under peak load and network stress.

While this architectural decoupling confers elastic scalability and operational flexibility, it creates a formidable **visibility barrier** for system performance, reliability, and computational sustainability:

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- **Indirect Failure and Degradation Pathways:** In traditional synchronous architectures (e.g., RESTful HTTP or gRPC), component interactions follow explicit caller–callee invocation paths. In asynchronous pub-sub and event meshes, publishers and subscribers share no direct references. Cascading failures, queue head-of-line blocking, and backpressure propagate across hidden logical paths spanning brokers, shared topics, colocated execution nodes, and shared libraries.
- **Distinct Degradation Mechanisms:** Disturbances in complex distributed systems do not propagate in a uniform manner. They manifest either as *sequential cascades* (e.g., a slow subscriber causing broker queue saturation and upstream backpressure) or as *simultaneous blast radii* (e.g., a shared runtime library crash, memory exhaustion, or host machine outage instantly disabling multiple colocated services). Conventional architectural diagrams and static call graphs fail to represent these multi-layer dependencies.

Addressing these architectural vulnerabilities is most effective and cost-efficient **prior to deployment**, during design and Continuous Integration / Continuous Delivery (CI/CD), adhering to established foundational principles of dependable computing [8]. However, at design and build time, **no runtime telemetry, distributed tracing, or operational logs exist**. Consequently, software architects, performance engineers, and Site Reliability Engineers (SREs) face two fundamental questions without operational data:

1. *Which components, message topics, and communication links are systemically critical to system dependability and performance?*
2. *Why are they critical, and what specific architectural repair (such as replicating a message broker, decoupling an over-subscribed topic, or sandboxing a shared library) will most effectively eliminate that risk?*

The same questions bear on computational sustainability, though we are careful about the form of the claim. Analyzing an architecture from a manifest requires no provisioned cluster, no running services, and no fault-injection harness — the resource it eliminates is infrastructure rather than CPU time. What it does not do is compute less: the deterministic structural analysis this framework depends on is expensive, and on our own corpus it costs substantially more than the discrete-event simulation it produces labels from (Sections 7.5 and 8.2). We measure wall-clock latency rather than energy counters, and we make no claim that the pipeline is computationally cheap in absolute terms. The narrow and supported statement is that the *learned* component is negligible within it — a 56 ms forward pass against minutes of feature extraction — so applying graph neural networks to pre-deployment analysis does not itself introduce a meaningful cost.

1.2. Problem Statement: The Architecture–Code Gap and the Black-Box AI Challenge

We formulate pre-deployment dependability and performance analysis around two distinct, complementary tasks:

1. **Failure-Impact Forecasting (Predictive Pathway) — the primary task.** We forecast dynamic cascading failure blast radii and identify critical components using a data-driven, relation-specific model over learned topological representations. Closed-form topological metrics capture broad connectivity cheaply; whether they also resolve multi-hop, relation-dependent cascade spread across heterogeneous channels is an empirical question, which we test directly against such a baseline (Section 7.1). Our predictive pathway is trained and evaluated against independent simulation ground truth as a ranking and critical-set identification model.
2. **Explainable Criticality Attribution (Explanation Layer) — what a rank alone cannot say.** A ranked shortlist indicates *where* risk lies, but not *how to fix it*. We therefore pair the predictor with an interpretable structural quality profile grounded in ISO/IEC 25010 [9] and ISO/IEC 25019 [10]. This layer diagnoses the *qualitative root cause* of vulnerability—distinguishing, for instance, an unreplicated single point of failure from a high-coupling maintainability bottleneck—to guide concrete repairs. It serves strictly as an attribution model, not a ranking model.

This separation is architectural rather than merely presentational: both pathways operate on the same graph but share no parameters, and neither is trained on the other’s output. The coupling term that could connect them is disabled by default and reported only as an ablation (Section 4.2). Maintaining this independence allows SaG to identify components that are structurally central yet operationally low-impact—a nuanced diagnosis unattainable by either pathway alone.

Existing software engineering approaches fail to bridge what we define as the **Architecture–Code Gap**: *a distributed system can have pristine, bug-free source code within each individual service, yet remain fragile to catastrophic global outages caused by hidden architectural single points of failure (SPOFs) or mismatched middleware Quality-of-Service (QoS) contracts*. Classical architecture evaluation such as ATAM [11, 12], and the literature on architectural technical debt [13] and bad smells [14], identify architectural risks but rely on manual stakeholder elicitation rather than quantitative structural analysis. The automated paradigms each leave a different part of the gap unaddressed: static code analysis [15, 16, 17, 18] cannot see message queues or cross-host propagation; chaos engineering [19] needs a provisioned cluster and arrives after the architecture is fixed; and homogeneous centrality [20, 21, 22, 23] flattens the system into an untyped graph in which a topic, a library and a host are indistinguishable. Section 2 develops each in turn.

Furthermore, while machine learning has demonstrated remarkable success across software engineering, contemporary AI approaches applied to system dependability often function as uninterpretable black boxes. Deep neural models frequently output scalar risk scores or latent embeddings without providing transparent, actionable rationales for their predictions. In mission-critical software engineering, an opaque risk score is inadequate: developers and SREs cannot refactor code or reconfigure infrastructure without understanding *why* a component is vulnerable and *which* architectural mechanism is compromised.

1.3. The Software-as-a-Graph (SaG) Approach

To bridge the Architecture–Code Gap while overcoming the black-box AI challenge, this work introduces **Software-as-a-Graph (SaG)**, an AI-driven pre-deployment **Static System Analysis (SSA)** framework. SaG ingests Architecture-as-Code manifests and executes a four-stage pipeline:

1. **Typed Multigraph Formulation**: SaG models the distributed architecture as a typed, directed multigraph over five core entity types: Applications, Brokers, Topics, Execution Nodes, and Shared Libraries (Section 3.1).
2. **QoS-Aware Logical Dependency Projection**: Using six formal projection rules, SaG derives a semantic `DEPENDS_ON` dependency layer that captures both sequential cascades (via topics and brokers) and simultaneous blast radii (via shared libraries and node colocation), weighted by declared QoS contracts (Section 3.2).
3. **Heterogeneous Graph Learning for Failure Forecasting (Predictive Pathway)**: SaG trains a **Heterogeneous Graph Transformer (HGT)** whose relation-specific attention lets a `USES` edge into a shared library propagate differently from a `PUBLISHES_TO` edge into a topic. It forecasts cascading blast radii, ranks critical components, and outputs per-relationship criticality alongside multi-task quality outputs (Section 4).
4. **Explainable Quality Attribution (Explanation Layer)**: To explain *why* a flagged component is critical, SaG combines code-level SCA metrics with topological properties into a deterministic **Reliability–Maintainability (RM)** attribution model (Section 5). Reliability decomposes into **Fault Tolerance** (error propagation depth) and **Availability** (single-point-of-failure exposure), pointing to distinct repairs. Because it is a linear, propagation-free aggregate by design, it explains *why* a component is vulnerable rather than how far a cascade travels; its standalone rank correlation is correspondingly modest (Section 7.1).

To ensure methodological rigor, SaG enforces a strict **input–label independence guarantee**: learned models and attribution baselines operate exclusively on the analytical graph G_{analysis} , while ground-truth failure impacts are generated by independent discrete-event simulators operating on the raw structural topology $G_{\text{structural}}$ (Section 4.4).

Figure 1 shows how the two pathways relate. The predictive pathway is the primary one and the only pathway validated against the simulation oracle — the oracle scores rankings, which a quality profile is

not — and that oracle is strictly an offline training-and-validation component, never a dependency of online inference. The explanation layer then characterizes what the predictor flagged and the remediation that implies. The single link between them is triage rather than data flow: the architect applies the explanation to whatever the predictor ranked. The remediation guidance closes a loop of its own, in which each candidate edit is re-simulated on its own mutated copy of $G_{\text{structural}}$ and kept only if it beats the simulator’s seed-to-seed noise.

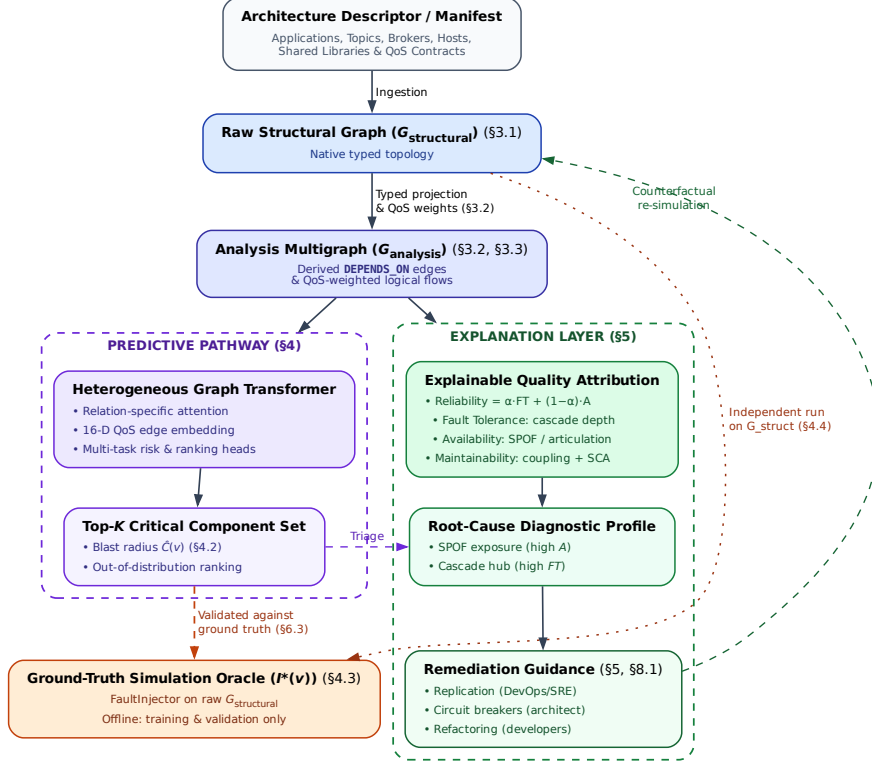


Figure 1: End-to-end architecture of the SaG framework. A shared front end (manifest ingestion → typed multigraph → QoS-weighted DEPENDS_ON projection → typed node features) feeds two pathways that share no parameters: the predictive pathway (Section 4), which emits a ranked critical set and per-relationship criticality, and the explanation layer (Section 5), which emits a standards-grounded quality profile. The simulation oracle scores only the former and runs on $G_{\text{structural}}$ alone.

Rationale for Graph Learning vs. Direct Simulation. Since discrete-event simulation $I^*(v)$ defines ground-truth criticality here, it is fair to ask why train a graph model at all rather than run simulation sweeps or closed-form heuristics. Three reasons motivate the design, and Section 7.1 tests them adversarially. A trained model scores entity types no simulator sweep was run for, since message passing generalizes across labeled and unlabeled entities alike. Cascade simulation is stochastic and seed-sensitive (label standard deviation reaches 0.416), whereas a trained model learns a smooth, threshold-marginalized surrogate that re-scores an already-analyzed architecture cheaply. And dynamic simulators need runnable containers or communication harnesses, whereas graph learning scores Architecture-as-Code manifests before any runtime infrastructure exists. A fourth motivation — that neither a simulator nor an unaugmented GNN returns a root cause in standardized quality terms — motivates the explanation layer rather than the predictor, and is taken up in Section 5. Whether these motivations are borne out empirically is a separate question, and Section 7.1 answers it only partly in the framework’s favor.

1.4. Research Questions

This empirical study investigates five research questions:

RQ1 (Predictive Efficacy): *How accurately does heterogeneous graph learning predict cascading failure impact and identify the critical component set, compared with traditional, non-learning network metrics?*

RQ2 (Value of Architectural Typing): *Does modeling distinct entity and dependency types (applications, topics, brokers, hosts, and libraries) yield better failure predictions than homogeneous graph models, and does that advantage hold on architectures the model has never seen?*

RQ3 (QoS Encoding and Robustness): *Do middleware Quality-of-Service contracts carry signal a purely structural score discards, do the framework’s simulation oracles agree with one another, and are the reported orderings robust to the free parameters of the scorer and of the ground truth?*

RQ4 (Real-World Generalization): *How effectively does the framework transfer zero-shot to authentic, real-world distributed systems across autonomous driving (ROS 2), cloud-native microservices, smart home IoT, and industrial edge computing?*

RQ5 (Analysis Cost): *What does pre-deployment analysis cost at CI/CD time, which pipeline stage dominates that footprint, and how does it compare against the discrete-event simulation it is intended to displace?*

1.5. Key Contributions

This paper presents four principal contributions:

- 1. Heterogeneous Graph Learning for Pre-Deployment Dependability:** A relation-specific Heterogeneous Graph Transformer that forecasts cascading blast radii from Architecture-as-Code manifests, with a 16-D edge feature vector carrying 7 QoS dimensions and multi-task heads for component and relationship criticality (Section 4). Under inductive distribution shift across twelve architectures, typed learning leads untyped learning by $\Delta\rho = +0.114$ in Spearman rank correlation (winning 11 of 12 folds, $p = 0.0122$), while in-distribution the two are indistinguishable (Section 7.2). The QoS edge encoding independently contributes $+0.054$ ($p = 0.0093$; Section 7.3.1). However, against an unparameterized QoS-weighted centrality baseline, out-of-distribution ranking is not statistically surpassed ($+0.127$, $p = 0.077$), establishing clear empirical boundaries (Section 7.1).
- 2. A Formal Typed Architecture Model:** A multigraph representation that derives logical dependencies from physical pub-sub linkages and distinguishes sequential cascade propagation from simultaneous multi-consumer library failures (Section 3).
- 3. A Standards-Grounded Explanation Layer:** An interpretable Reliability–Maintainability model grounded in ISO/IEC 25010/25019 that turns ranked predictions into actionable diagnoses, separating single-point-of-failure exposure from error-propagation reach (Section 5).
- 4. Empirical Benchmark, Real-World Transfer, and Cost Profile:** An evaluation across twelve synthetic topologies (2,461 components) and five open-source systems (351 components) under strict graph-view separation. We characterize pipeline cost, showing that the neural model represents only 0.02% of runtime, while deterministic feature analysis dominates (82.7 s vs. 7.2 s for simulation), leading us to withdraw the conference version’s computational-efficiency claim (Sections 6–7).

Relationship to the authors’ prior work. An earlier conference paper [24] introduced the preliminary multi-graph formulation and deterministic quality model on synthetic topologies. This JSS manuscript substantially extends that work by introducing: the entire predictive HGT pathway with 16-D QoS edge encoding and multi-task heads (Section 4); inductive LOSO cross-validation (Section 7.2); zero-shot evaluation across five open-source systems (Section 7.4); empirical cost and sustainability characterization (Section 7.5); multi-oracle convergent validity and graph-view separation (Sections 4.3–4.4); and global sensitivity analyses (Section 7.3). Retained formalisms from the conference paper are limited to restructured portions of Sections 3 and 5.

1.6. Paper Organization

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 formalizes the SaG multigraph model and dependency projections. Section 4 details the Heterogeneous Graph Transformer and simulation oracles. Section 5 presents the ISO/IEC-grounded explanation layer. Section 6 outlines the experimental methodology, while Section 7 reports empirical results for RQ1–RQ5. Section 8 discusses practical implications, sustainability, threats to validity, and limitations. Section 9 concludes.

2. Related Work

This work builds upon and connects four foundational research areas: (1) dependability, performance, and sustainability in distributed software systems; (2) static code and system analysis; (3) software quality measurement and multi-criteria evaluation; and (4) graph representation learning and explainable AI (XAI).

2.1. Dependability, Performance, and Sustainability in Distributed Software Systems

The publish–subscribe (pub-sub) and asynchronous event-driven paradigms decouple communicating entities in space, time, and synchronization, enabling elastic scalability and high throughput [7]. Modern middleware standards—such as ROS 2 [1], Apache Kafka [2], DDS [3], and MQTT [4]—govern these exchanges through fine-grained Quality-of-Service (QoS) policies that regulate message durability, transport reliability, priorities, and delivery deadlines. In cloud-native microservice meshes and distributed AI/LLM serving backbones, asynchronous message passing and queueing topologies form the primary communication substrate, directly shaping tail latencies, throughput bottlenecks, and hardware resource utilization.

Prior dependability and performance research has focused predominantly on **runtime mechanisms**, including dynamic consensus protocols, broker clustering, adaptive backpressure throttling, autoscaling, and automated failover. In parallel, **chaos engineering and runtime verification** [19] inject faults or latency into staging or production clusters to observe degradation and recovery. While runtime fault injection delivers operational validation that no static method can match, it requires a fully provisioned cluster, carries the risk of real service disruption, and consumes cluster-hours per sweep — which places it, alongside model training, among the development-time computations whose energy cost green software engineering has argued should be accounted for rather than assumed away [25, 26, 27]. In practice this precludes its use during architectural design or lightweight commit-level CI/CD.

Our work addresses the complementary **pre-deployment phase**: predicting systemic cascading vulnerabilities and performance degradation directly from Architecture-as-Code descriptors before runtime infrastructure is provisioned. From a green software engineering perspective, the input is a manifest rather than an active deployment. We are careful about how far this argument reaches: it is a claim about what must be provisioned, not that the analysis uses less computation than alternatives—a distinction our measurements force, since the static gate proves more expensive than the simulation oracle it was intended to displace (Sections 7.5.1 and 8.2). Avoiding production restart storms remains motivating rather than an empirical claim.

Architecture-Based Reliability Prediction. Predicting dependability from an architectural description before deployment is not a new ambition, and SaG should be read against the tradition that pursued it analytically. Cheung’s absorbing-Markov-chain model [28] derives system reliability from component reliabilities and a transfer-of-control graph; Goseva-Popstojanova and Trivedi [29] systematize the state-based, path-based and additive families that followed, and Immonen and Niemelä [30] survey the resulting methods from the architectural perspective. Model-driven descendants such as the Palladio Component Model [31] and layered queueing networks [32] predict performance and reliability from parameterized component models with well-understood solution techniques.

A parallel tradition explicitly annotates architectural descriptions with fault behaviors or parameters. For example, the AADL Error Model Annex [33] allows architects to declare component error states and propagation paths to generate fault trees and Markov models automatically. Similarly, analytical reliability models require per-component failure probabilities, transition rates, or operational profiles [29, 31]. Where

authored, such models answer strictly richer questions than ours. The critical distinction lies in required inputs: analytical and annex-based methods require deliberate, pre-calibrated failure semantics unavailable at commit time without operational telemetry. SaG asks a narrower question in exchange: given only declared deployment manifests, which components’ failures would propagate furthest through the declared topology? Where those parameters can be obtained, an analytical model answers a stronger question than a ranking does, and we make no claim to displace it.

Data-Driven Failure Prediction and Root-Cause Analysis in Microservices. A large recent literature localizes faults in microservice systems from operational data. Seer [34] and Sage [35] predict and debug QoS violations from traces and hardware telemetry; MicroRCA [36] and TraceRCA [37] localize root causes over service-dependency and trace graphs; DeepTraLog [38] and Eadro [39] combine traces, logs and metrics under graph-based deep models. This line is the closest methodological neighbor to our predictive pathway, and it consistently outperforms what a purely static analysis can achieve — because it observes the running system. That is precisely the boundary: every one of these approaches requires a deployed system emitting traces, logs or metrics, and therefore cannot answer a question posed at design or pull-request time. SaG occupies the pre-deployment complement, and accepts a correspondingly weaker evidential basis: simulated rather than observed failures, and topology rather than behavior.

2.2. Static Code Analysis (SCA) vs. Static System Analysis (SSA)

Traditional **Static Code Analysis (SCA)** tools (e.g., SonarQube [15]) inspect source code Abstract Syntax Trees (ASTs) within individual services. They evaluate cyclomatic complexity [16], class cohesion, module coupling (e.g., Lack of Cohesion in Methods [LCOM], Coupling Between Objects [CBO]) [17, 18], and code duplication to flag internal code smells and defect-prone modules [40, 41, 42, 43]. However, SCA cannot observe runtime communication topology: it is blind to inter-service messaging channels, message broker queue saturation, and cross-host failure propagation.

Recovering system-level structure statically is, however, an active area in its own right, and we do not claim the idea as novel. A body of work reconstructs microservice architecture from source and deployment artifacts without running the system: Bushong et al. [44] derive communication diagrams and bounded contexts from static code analysis of a service mesh, and a recent multivocal review compares nine such recovery tools and finds their outputs complementary enough that combining them improves detection [45]. That literature and ours differ in what the recovered graph is *for*: architecture recovery aims to reproduce a faithful description of the system as built, typically for comprehension or drift detection, whereas we take a declared topology as given and ask which of its components a failure would propagate furthest from. Recovery is, in that sense, an upstream complement — it could supply the manifests SaG consumes for a system whose Architecture-as-Code description is incomplete.

To bridge this “Architecture–Code Gap,” **Static System Analysis (SSA)** extends static analysis from single-service source code to the global system architecture. By modeling distributed applications, message topics, brokers, execution nodes, and shared libraries as a connected multigraph, SSA propagates code-level quality metrics across architectural dependencies. This allows engineering teams to detect structural anti-patterns [46, 47] and architectural technical debt [48] early during continuous integration (CI/CD) [49, 50], before defective topologies enter production.

2.3. Software Quality Models and Multi-Criteria Evaluation

Software product quality is standardized by the **ISO/IEC 25010:2023** product quality model [9] and the **ISO/IEC 25019:2023** Quality-in-Use model [10]. ISO/IEC 25010:2023 defines three closely intertwined characteristics critical to modern distributed systems:

- **Reliability:** The degree to which a system performs specified functions under stated conditions, comprising Faultlessness, Availability, Fault Tolerance, and Recoverability.
- **Maintainability:** The degree of effectiveness and efficiency with which software can be modified, comprising Modularity, Reusability, Analyzability, Modifiability, and Testability.

- **Performance Efficiency:** Performance relative to resource consumption under stated conditions, comprising Time Behavior (latency, response time), Resource Utilization (CPU, memory, bandwidth), and Capacity.

SaG operationalizes a strict subset of these: Availability and Fault Tolerance under Reliability, and Modularity, Modifiability, and Analyzability under Maintainability (Section 5.1). Faultlessness, Recoverability, Reusability, and Testability are not derivable from deployment topology alone and are outside the scope of this work.

Software engineering measurement explicitly distinguishes between *internal quality* (measured on static artifacts at rest) and *external quality* (measured on executing software systems) [51, 52]. In distributed architectures, architectural debt (such as over-centralized message topics or unreplicated brokers) degrades internal quality and precipitates severe external performance bottlenecks, queue congestion, and outages.

Aggregating multi-attribute structural metrics into an auditable quality score constitutes a classic Multi-Criteria Decision Making (MCDM) problem. The **Analytic Hierarchy Process (AHP)** [53] delivers a structured pairwise-comparison method with an explicit Consistency Ratio ($CR \leq 0.10$) intended to certify that elicited judgments are mutually coherent. That statistic detects *inconsistency*; it cannot detect a matrix filled in from an answer already chosen, which is a limitation we take seriously for our own weights and quantify in Supplementary Section S4. This study applies AHP to construct an audited, explainable Reliability–Maintainability (RM) quality baseline, in conjunction with learned graph models.

2.4. Graph Representation Learning and Explainable AI

Network science provides established centrality metrics to identify critical nodes, including degree, closeness, betweenness centrality [20, 22], articulation points, and PageRank [21, 23]. Foundational studies on network robustness [54], cascading overloads [55], and interdependent networks [56] model disruption propagation across connected topologies. While percolation models offer natural comparators, our training-free baselines are centrality-based (Section 6.2); evaluating targeted percolation fragmentation remains a recognized future baseline comparison.

However, standard network metrics suffer from two major limitations when applied to software architectures: (1) **Dimensional Collapse**, where a single centrality scalar cannot distinguish *why* a component is critical (e.g., an isolated single point of failure vs. an error-propagating cascade hub vs. an over-shared library); and (2) **Semantic Collapse**, where unweighted metrics treat all nodes and edges identically, conflating fundamentally different architectural entities such as asynchronous message topics, shared libraries, and physical execution hosts.

To overcome hand-engineered metrics, recent studies apply machine learning to network vulnerability (e.g., FINDER [57], DrBC [58], PowerGraph [59]). However, most models rely on **homogeneous message passing** (GCN [60], GraphSAGE [61], GAT [62]), averaging signals indiscriminately across connection types. Because distributed software architectures are inherently **heterogeneous**, homogeneous models blur entity boundaries and fail to generalize out-of-distribution. Heterogeneous Graph Neural Networks (RGCN [63], HAN [64], HGT [65], MAGNN [66]) resolve this via relation-specific transformations. We build upon the **Heterogeneous Graph Transformer (HGT)** [65] to preserve typed relational semantics when forecasting cascade blast radii.

Explainable AI (XAI) vs. The Black-Box Barrier. A critical hurdle in applying modern AI to software engineering is the **black-box barrier**: deep neural models output risk scores or continuous embeddings without explaining underlying structural causality. In production software engineering, uninterpretable risk rankings hinder actionable decision-making: developers and SREs cannot determine whether to replicate a host, configure circuit breakers, or refactor shared libraries.

Existing GNN explanation techniques, such as GNNExplainer [67] and PGExplainer [68], identify influential subgraphs through edge masking or parameterized learning. Although useful, these methods explain the model using internal latent representations rather than standardized software engineering concepts. SaG resolves this limitation through a decoupled dual-pathway design: the predictive HGT pathway reveals typed mutual-attention distributions indicating *which* architectural relations propagated the

cascade (Section 7.3.4 and Supplementary Section S8), while the deterministic explanation layer attributes fragility to standardized ISO/IEC quality sub-characteristics (Section 5), translating raw predictions into actionable, cost-effective remediations.

3. The Software-as-a-Graph (SaG) Architectural Model

This section formalizes the Software-as-a-Graph multigraph representation (Section 3.1), the QoS-aware weighting and logical dependency derivation rules (Section 3.2), the dual graph views (Section 3.3), and the typed node feature encodings (Section 3.4).

3.1. Formal Multigraph Definition

A complex distributed software system is formally modeled as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- V is the set of system entities, partitioned into five disjoint entity types:

$$V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{node}} \cup V_{\text{lib}} \quad (2)$$

- E is the set of directed edges connecting entities.
- $\tau_V : V \rightarrow \mathcal{T}_V$ and $\tau_E : E \rightarrow \mathcal{T}_E$ are typing functions assigning node and edge categories.
- $w_V : V \rightarrow [0, 1]$ and $w_E : E \rightarrow [0, 1]$ are weighting functions representing entity criticality and connection strength.

Table 1 summarizes the five entity types and six structural edge types formalized in the SaG model, along with their semantics and representative concrete distributed-system implementations.

Table 1: Entity and structural edge types in the SaG model.

Entity Type ($\mathcal{T}_V$)	Architectural Role	Concrete System Examples
Application (V_{app})	Autonomous process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	/sensor/lidar, orders.payment.completed
Node (V_{node})	Physical host or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	librdkafka, OpenCV, Protobuf runtime
Structural Edge ($\mathcal{T}_E$)	Direction	Semantic Meaning
PUBLISHES_TO	App/Library $\rightarrow$ Topic	Component publishes messages to topic
SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component consumes messages from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Node	Process is hosted on physical/virtual host
CONNECTS_TO	Node $\rightarrow$ Node	Physical network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library dependency

Application and Library entities additionally incorporate static code metrics computed via Static Code Analysis (SCA) tools (`cm_*` attributes: lines of code, cyclomatic complexity, coupling between objects, LCOM), linking code-level fragility directly to topological analysis.

3.2. QoS-Aware Weights and Logical Dependency Derivation

In distributed middleware, communication links vary in coupling strength based on their Quality-of-Service (QoS) contracts. For instance, a **RELIABLE** topic with **TRANSIENT_LOCAL** durability binds communicating services substantially more tightly than a **BEST_EFFORT** telemetry stream.

Each topic t carries an intrinsic criticality weight $w(t) \in [0, 1]$ combining its declared QoS semantics with two runtime-stress modulators: payload size and publication frequency:

$$w(t) = \beta \cdot \text{QoS}(t) + \alpha \cdot \text{SizeNorm}(t) + \psi \cdot \text{FreqNorm}(t), \quad (\beta, \alpha, \psi) = (0.75, 0.15, 0.10) \quad (3)$$

where the QoS term is an AHP-weighted aggregate of the declared contract:

$$\text{QoS}(t) = w_{\text{rel}} \cdot q_{\text{rel}} + w_{\text{dur}} \cdot q_{\text{dur}} + w_{\text{prio}} \cdot q_{\text{prio}}, \quad (w_{\text{rel}}, w_{\text{dur}}, w_{\text{prio}}) = (0.24, 0.62, 0.14) \quad (4)$$

Here, $q_{\text{rel}}, q_{\text{dur}}, q_{\text{prio}} \in [0, 1]$ represent normalized reliability, durability, and transport-priority scores. Durability dominates because it governs whether data persists across restarts and network partitions. Reliability and transport priority both govern in-flight delivery quality, with reliability receiving higher weight because unconditional delivery guarantees precede message scheduling. The sub-weight vector is the geometric-mean priority vector of an independently stated Saaty pairwise-comparison matrix, printed with its consistency computation in Supplementary Section S4; $CR = 0.016$, small but non-zero. The modulators are logarithmically compressed and clamped to $[0, 1]$: $\text{SizeNorm}(t) = \min(1.0, \log_2(1 + \text{bytes})/20)$ (a 1 MiB design envelope, representing the practical DDS sample ceiling before RTPS fragmentation dominates) and $\text{FreqNorm}(t) = \min(1.0, \log_{10}(1 + \text{Hz})/3)$. The final weight $w(t)$ is clamped to $[0.01, 1]$, ensuring that best-effort edges remain visible to graph traversals. Every structural communication edge incident on t (**PUBLISHES_TO**, **SUBSCRIBES_TO**, **ROUTES**) inherits $w_E(e) = w(t)$ along with the topic's QoS vector.

The outer split (β, α, ψ) is a declared convex combination. Sweeping it over the full simplex changes the induced ordering of $w(t)$ by at most $\rho = 0.076$ and downstream rank correlation by at most 0.020, so it is a documented convention rather than a tuned parameter (Supplementary Section S1).

3.2.1. Logical Dependency Projection (**DEPENDS_ON**)

Structural edges capture explicit deployment connections but omit implicit runtime dependencies. For example, a subscriber depends upon a publisher, yet no direct edge connects them in pub-sub architectures. We therefore derive a single unified semantic relation, **DEPENDS_ON**, directed from *dependent* to *dependency* (“if target fails, source is impacted”), according to the six projection rules detailed in Table 2:

Table 2: The six **DEPENDS_ON** logical dependency projection rules.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic, incl. transitive USES)	$1 - \prod_{t \in T} (1 - w(t))$
2	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3	node_to_node	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	Lifted max w
4	node_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	Lifted max w
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6	broker_to_broker	Broker $\leftrightarrow$ Broker (shared physical fault-domain colocation, symmetric)	$w_V(\text{node})$

Rules 1 and 2 aggregate the set of topics T connecting a component pair using a probabilistic union rather than a maximum [69, 70, 71]. This guarantees that additional parallel failure vectors increase coupling monotonically while keeping $w \in (0, 1]$. Rule 5 applies the harmonic mean $H(x, y) = 2xy/(x + y)$ [72] to combine the consuming Application's and the shared Library's vertex weights, balancing caller and dependency criticality. Rules 3 and 4 assign the maximum weight among component-level dependencies crossing the host boundary.

3.2.2. Sequential Cascades vs. Simultaneous Blasts

A foundational principle of the SaG model is distinguishing between two fundamentally different degradation modes:

- **Sequential Cascade (Rule 1):** When an application publisher fails, downstream subscribers suffer message starvation. The failure propagates hop by hop through message queues and topic buffers.
- **Simultaneous Blast (Rule 5):** When a shared software library or execution node crashes, all consuming applications and colocated brokers fail *instantaneously* in a single shared-fate event.

Preserving architectural entity types and relation-specific projection rules enables SaG to model both mechanisms, whereas untyped homogeneous graphs collapse them into indistinguishable edges.

Rule 6 is intentionally the only symmetric projection rule. It does not imply that one broker functionally depends on another, but rather that two brokers colocated on the same host share that host’s physical failure domain. This follows the same simultaneous-blast principle as Rule 5, which is why the derived weight equals the shared Node’s weight and the relation is bidirectional. In production middleware deployments, colocated brokers compete for host resources (CPU cores, page cache, file descriptors, and NIC bandwidth); a host outage takes down all colocated instances simultaneously. Operational best practices for Kafka, RabbitMQ, and EMQX recommend distributing brokers across fault domains. Rule 6 does not model directional intra-cluster broker coupling (e.g., partition replication, controller quorum election, federation, or shovel links), which do not require physical colocation; extending the schema to capture these interactions is reserved for future work. Rule 6 applies in four of the eight scenarios forming the detection benchmark subset — the seven core synthetic topologies plus the ATM case study, the subset identified in Table 5 — and contributes only 12 directed edges across them. Because the simulation oracles operate strictly on $G_{\text{structural}}$ (Section 4.4), Rule 6 has zero influence on ground-truth failure labels.

3.3. Dual Graph Views and Architectural Layers

The SaG framework maintains two distinct representations of the system:

1. **Structural Graph ($G_{\text{structural}}$):** The raw deployment graph containing physical and structural relations (such as PUBLISHES_TO, ROUTES, RUNS_ON, and USES). Discrete-event simulators consume this view exclusively to execute unbiased failure injections (Section 4.3).
2. **Analysis Graph (G_{analysis}):** The projected graph containing derived DEPENDS_ON edges annotated with QoS weights and ingested SCA code metrics. All GNN feature representations, graph embeddings, and analytical metrics are computed on G_{analysis} .

Figure 2 illustrates this duality on a running example, contrasting the raw structural graph against the derived DEPENDS_ON projection.

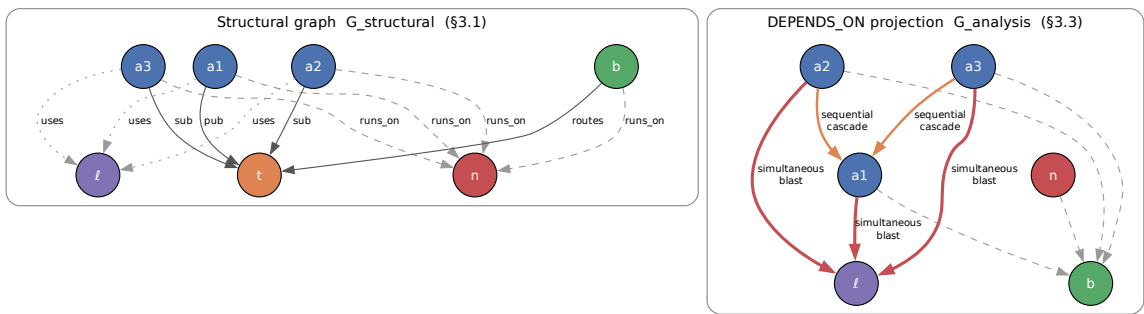


Figure 2: Running example: the raw structural graph (left) and the DEPENDS_ON projection derived from it (right). The projection makes implicit runtime dependencies explicit—a subscriber depends on the publishers of its topics even though no structural edge joins them—while the simulators continue to operate on the structural view alone.

G_{analysis} is further structured into four analytical layers (Application, Middleware, Infrastructure, and Global System), enabling evaluation of criticality at subsystem levels, consistent with hierarchical frameworks such as MIL-STD-498 [73].

3.4. Typed Node Feature Encoding

Both pathways read the same typed node properties from G_{analysis} : the predictive pathway (Section 4) projects them per entity type before heterogeneous message passing, and the explanation layer (Section 5) aggregates them into its quality profile. All five entity types share indices 0–17, a common block of topological metrics (in/out degree, betweenness, closeness, reverse PageRank, clustering coefficient, articulation score, bridge load) produced by the deterministic analysis stage whose cost is characterized in Section 7.5. Type-specific features extend that block:

- **Application (23 dims):** indices 18–22 add source-code metrics from SCA — lines of code, cyclomatic complexity, Martin’s instability $I_{\text{code}} = C_e/(C_a + C_e)$ [74], Lack of Cohesion in Methods, and the composite Code Quality Penalty (CQP).
- **Library (25 dims):** the Application block plus two library-specific blast-radius drivers (23–24): the normalized size of the transitive reverse-USES closure, and the normalized count of distinct subscribers reachable from topics published within that closure.
- **Broker (19 dims):** index 18 is normalized queue buffer capacity.
- **Topic (22 dims):** indices 18–21 are publisher count, subscriber count, log message frequency $\log(1 + \text{freq})$, and ordinal QoS criticality.
- **Infrastructure Node (20 dims):** indices 18–19 are normalized CPU core allocation and physical memory.

4. Graph Learning for Failure-Impact Prediction

Cascading failure impact in distributed software systems is inherently non-linear, multi-hop, and relation-dependent. Outages propagate not merely based on neighbor count, but through architectural relations and dependencies extending multiple hops beyond the initial fault. Whether a closed-form combination of standard centrality metrics can capture these compound dynamics is an empirical question rather than a settled one. The primary predictive pathway of Section 1.2 therefore employs a learned graph model, and Section 7.1 evaluates it against exactly such a closed-form baseline — which, on out-of-distribution ranking, it does not significantly surpass.

This section details the Heterogeneous Graph Transformer (HGT) architecture and its typed edge encodings (Section 4.1), the multi-task prediction heads and dimension-masked loss formulation (Section 4.2), the ground-truth simulation oracles (Section 4.3), and the input-label independence guarantee that prevents data leakage (Section 4.4).

4.1. Heterogeneous Graph Transformer Architecture

Because distributed systems comprise heterogeneous entity types (Applications, Libraries, Brokers, Topics, Infrastructure Nodes) and diverse interaction semantics (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON), we employ a three-layer **Heterogeneous Graph Transformer (HGT)** architecture [65], implemented within PyTorch Geometric [75], with hidden dimension $D = 64$ and $H = 4$ attention heads. This architecture ensures that typed relations, rather than simple adjacency, govern failure-impact forecasting.

4.1.1. Continuous-Categorical Edge Feature Encoding (16-D)

To capture continuous QoS constraints and channel semantics, SaG encodes each directed edge $e = (u, v)$ as a 16-dimensional continuous-categorical vector $e_{uv} \in \mathbb{R}^{16}$. Index 0 is the scalar coupling weight $w_E(e) \in (0, 1]$ of Section 3.2; index 1 is the normalized count of simple paths through e in G_{analysis} ; indices 2–8 one-hot encode the seven structural and derived relations (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON). Indices 9–15 encode middleware QoS parameters on PUBLISHES_TO and SUBSCRIBES_TO edges (zeroed elsewhere). Four of these dimensions are active in our evaluation corpus:

reliability (0 best-effort, 1 reliable); durability (0 volatile, 0.5 transient-local, 0.6 transient, 1 persistent); message priority (0, 0.33, 0.66, 1); and a heterogeneity flag set when the edge’s QoS triple departs from its scenario’s modal profile. The remaining three indices (deadline-active flag, $\log_{10}(1 + \text{deadline_ns}/10^6)$, and $\log_{10}(1 + \text{max_blocking_ms})$) are architectural schema provisions designed to accommodate hard real-time DDS and ROS 2 microsecond profiles, and are zero-initialized in the current benchmarks.

An edge projection module maps e_{uv} into the hidden space: $e'_{uv} = W_{\text{edge}}e_{uv}$. Prior to relational attention computation, this projection vector is incorporated directly into the target node representation: $\tilde{h}_v = h_v + e'_{uv}$.

4.1.2. Type-Specific Projection and Heterogeneous Message Passing

For each source node u and target node v connected by meta-relation $\tau(e) = (\tau(u), \phi(e), \tau(v))$:

1. **Type-Specific Projection:** Node feature vectors x_v (of dimension 19–25 depending on entity type $\tau(v)$) are mapped into the shared D -dimensional hidden space:

$$h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v)) \quad (5)$$

2. **Relational Mutual Attention:** Type-parameterized Query (Q), Key (K), and Value (V) projections calculate relation-specific attention. For head $i \in \{1, \dots, H\}$, with the softmax taken over the incoming neighborhood $\mathcal{N}(v)$:

$$\text{Attn}^i(u, e, v) = \text{Softmax}_{u \in \mathcal{N}(v)} \left(K^i(u) W_{\text{att}, \phi(e)}^i Q^i(\tilde{h}_v)^\top \cdot \frac{\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}}{\sqrt{D/H}} \right) \quad (6)$$

where $\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}$ is the learned per-meta-relation scaling prior of Hu et al. [65], which lets the model weight an entire relation triple up or down independently of the node pair. We retain it: it is the parameter that most directly expresses “this relation type matters more than that one”, and the typing effect of Section 7.2 is what it exists to capture. The implementation is PyTorch Geometric’s HGTCConv [75], whose `p_rel` parameter is this term.

$$\text{Msg}(u, e, v) = V(u)W_{\text{msg}, \phi(e)} \quad (7)$$

3. **Bidirectional Message Passing:** To capture downstream consumer starvation and upstream back-pressure simultaneously, message passing is executed over both forward and transposed relation views (G_{analysis} and $G_{\text{analysis}}^\top$).
4. **Residual Aggregation and Layer Normalization:** Target node representations are updated across layers $l \in \{1, \dots, L\}$ via residual connections and layer normalization:

$$h_v^{(l)} = \text{LayerNorm} \left(h_v^{(l-1)} + \text{Dropout} \left(\sum_{u \in \mathcal{N}(v)} \text{Attn}(u, e, v) \cdot \text{Msg}(u, e, v) \right) \right) \quad (8)$$

Training Protocol and Optimization Hyperparameters. Models are optimized end-to-end using AdamW with initial learning rate $\eta = 3 \times 10^{-4}$, weight decay 10^{-4} , and dropout probability $p = 0.10$ applied post-attention. Learning rates follow a cosine annealing schedule with warm restarts (CosineAnnealingWarmRestarts, $T_0 = 75$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 3 \times 10^{-6}$). Training executes for a maximum of 300 epochs with early stopping governed by a patience of 30 epochs monitored on validation loss over labeled nodes. Inductive subgraphs are processed per scenario using full-graph inductive packing without mini-batch subsampling, with validation masks isolating held-out nodes to prevent information leakage across data splits. Five independent random seeds {42, 123, 456, 789, 2024} are evaluated across all runs, redrawing both partition masks and initializations. *Selection protocol:* the architectural hyperparameters ($D = 64$, $H = 4$, dropout, learning rate, and schedule) follow values conventional for HGT [65]. The loss coefficients of Equation (9) have no such precedent — the objective is bespoke to this task — and were set by judgment and left untuned; we state this rather than appeal to a convention that does not exist for a five-term multi-task loss. Neither group was tuned against

the in-distribution test split or the LOSO folds; no search over them was performed there. The real-world evaluation of Section 7.4.1 is a separate case and is documented separately: it runs at a different depth and epoch budget from every other learned result in this paper, and Section 7.4.1 states that configuration and how it was arrived at. This avoids selection leakage, at the cost of leaving open whether either family is reported near its own optimum — a comparison between untuned configurations, which we state rather than treat as a like-for-like optimum comparison.

4.2. Multi-Task Prediction Heads and Dimension Masking

From the final node embeddings $h_v^{(L)}$, SaG utilizes specialized multi-task prediction heads:

- **Reliability Head:** $\hat{R}(v) = \sigma(\text{MLP}_R(h_v)) \in [0, 1]$
- **Maintainability Head:** $\hat{M}(v) = \sigma(\text{MLP}_M(h_v)) \in [0, 1]$
- **Composite Failure Impact Head:** $\hat{I}^*(v) = \sigma(\text{MLP}_C(h_v \parallel \hat{R}(v) \parallel \hat{M}(v))) \in [0, 1]$
- **Relationship Criticality Head:** $\hat{Q}(u, v) = \sigma(\text{TypedEdgeEncoder}_{\phi(e)}(h_u, h_v, e_{uv})) \in [0, 1]$

4.2.1. Dimension-Masked Loss Formulation

The combined optimization objective integrates regression accuracy, multi-task dimension learning, ranking fidelity, pairwise ordering, and edge prediction:

$$\mathcal{L} = \mathcal{L}_{\text{composite}} + 0.5 \cdot \mathcal{L}_{\text{dimension}} + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} + 0.3 \cdot \mathcal{L}_{\text{edge}} + \lambda_{\text{RM}} \cdot \mathcal{L}_{\text{consistency}} \quad (9)$$

where $I^*(v)$ is the simulated cascade impact defined by the primary oracle (Section 4.3), $\mathcal{L}_{\text{composite}} = \text{MSE}(\hat{I}^*(v), I^*(v))$, $\mathcal{L}_{\text{rank}}$ is the ListMLE ranking loss [76], $\mathcal{L}_{\text{pairwise}}$ is margin-ranking loss, and $\mathcal{L}_{\text{consistency}} = \text{MSE}([\hat{R}(v), \hat{M}(v)]_{v \in \text{unlabeled}}, [R_{\text{RM}}(v), M_{\text{RM}}(v)]_{v \in \text{unlabeled}})$ regresses predicted heads toward the diagnostic pathway’s baseline (Section 5) on unlabeled nodes. Headline results use $\lambda_{\text{RM}} = 0$, guaranteeing that the predictive and explanatory pathways remain strictly independent.

Dimension Masking: Because dynamic cascade simulation ($I^*(v)$ via **FaultInjector**) observes runtime failure reachability rather than source-code maintainability, maintainability ground truth is unobserved during dynamic simulation. A separate change-propagation oracle $I_M(v)$ evaluates static structural change ripple at the Validate stage, but is never used as a training label to avoid circular supervision. We introduce a boolean dimension mask $m = [m_R, m_M] = [1, 0]$:

$$\mathcal{L}_{\text{dimension}} = \frac{1}{\sum_d m_d} \sum_{d \in \{R, M\}} m_d \cdot \text{MSE}(\hat{d}(v), d^*(v)) \quad (10)$$

This mask ensures the unobserved maintainability head is not artificially penalized or driven toward zero during backpropagation.

What the reliability label is. The surviving term deserves to be stated plainly, because it is weaker than a multi-task objective normally implies. **FaultInjector** emits a single scalar per component, and the label extractor assigns that same scalar to both the composite and the reliability columns: $R^*(v) = I^*(v)$ identically. $\mathcal{L}_{\text{dimension}}$ under $m = [1, 0]$ therefore regresses $\hat{R}$ toward the very target $\mathcal{L}_{\text{composite}}$ regresses $\hat{I}^*$ toward. The two terms are not redundant — they train separate heads, and $\hat{R}$ re-enters the composite head as an input ($\hat{I}^* = \sigma(\text{MLP}_C(h_v \parallel \hat{R} \parallel \hat{M}))$), so the term acts as an auxiliary path to the same supervision rather than as a second source of information. But no dimension of the label is decomposed by this oracle: a reliability score distinct from overall impact would require an oracle that separates fault-tolerance from availability, which $I^*(v)$ does not. We report the objective as implemented rather than describing it as multi-dimensional supervision it does not receive.

4.2.2. Domain-Reweighted Criticality (Q_{domain})

To ground predictions in ISO/IEC 25019 Context of Use ($\vec{\omega} = [q_R, q_M]^\top$), the composite score can be evaluated as:

$$Q_{\text{domain}}(v) = q_R \cdot \hat{R}(v) + q_M \cdot M_{\text{static}}(v) \quad (11)$$

where $M_{\text{static}}(v)$ is obtained directly from the structural analyzer’s maintainability baseline, combining learned dynamic reliability with static source-code maintainability. Because maintainability is unobserved in dynamic simulation ($m = [1, 0]$), headline results report $\hat{I}^*(v)$ directly, while domain reweighting sensitivity is evaluated against the static RM baseline in Supplementary Section S3.

4.3. Ground-Truth Simulation Oracles

To evaluate predictive accuracy prior to deployment without relying on production runtime telemetry, SaG executes discrete-event failure simulations over the raw structural multigraph $G_{\text{structural}}$. We establish a formal taxonomy of four component-level oracles and one relationship-level oracle:

- **Cascade Reachability Oracle ($I^*(v)$)**, via **FaultInjector**: crashes component v , propagates outages across dependent topics, brokers, and links by breadth-first traversal, and returns the mean fractional feed loss over the subscriber population, each subscriber contributing the unweighted mean loss of the topics it subscribes to. A topic’s feed loss is the fraction of its publishers that have failed (for a topic with no publisher, the fraction of its failed routers), scaled by a QoS ladder ($\times 1.2$ RELIABLE, $\times 1.15$ high/urgent priority, $\times 1.05$ medium) and clamped to $[0, 1]$. The denominator is the full subscriber set of the intact graph, so subscribers that themselves fail are retained in the average rather than excluded. The implementation admits a per-publisher rate weighting, but rates are declared per topic throughout our corpus, so it reduces exactly to this publisher fraction. This is the **primary continuous target label** throughout.

- **Multi-Metric Composite Oracle ($I_{\text{comp}}(v)$)**, via **FailureSimulator**:

$$I_{\text{comp}}(v) = 0.35 \cdot \Delta\text{Reachability} + 0.25 \cdot \Delta\text{Fragmentation} + 0.25 \cdot \Delta\text{Throughput} + 0.15 \cdot \Delta\text{FlowDisruption} \quad (12)$$

each term weighted by operational severity $s(t) = w(t) \cdot \text{rate}(t)$. These four coefficients are AHP-derived on the same footing as the rest of the framework’s weights, from a Saaty pairwise-comparison matrix over the four impact criteria (Supplementary Section S4): the raw priority vector is $(0.389, 0.255, 0.255, 0.100)$, and the $\lambda = 0.7$ shrinkage toward a uniform prior that the framework applies throughout yields $(0.347, 0.254, 0.254, 0.145)$, which the shipped constants round. That matrix is rank-one, so it records where the constants came from without independently justifying them. They are not fitted to any evaluation, and they are not swept in the supplementary sensitivity analysis — a gap we note because I_{comp} supplies the labels for the real-world evaluation of the explanation layer (Supplementary Section S7). $\text{rate}(t)$ reads from a runtime telemetry profile when one is attached; none is attached anywhere in this paper, so $s(t) = w(t)$ throughout and the rate channel should be read as an interface for deployment-time calibration rather than an active term. Publication frequency and payload size still reach $s(t)$ through $w(t)$ ’s own modulators (Section 3.2).

- **Dynamic Queue-Flow Oracle ($I_{\text{dyn}}(v)$)**, via **MessageFlowSimulator** on SimPy [77]: simulates emission rates, stochastic latencies, broker buffer saturation, and queue drops under fault injection, extracting the drop in delivered message rate to surviving consumers.
- **Change-Propagation Oracle ($I_M(v)$)**, via **ChangePropagationSimulator**: a deterministic reverse-dependency traversal quantifying maintenance change impact,

$$I_M(v) = 0.45 \text{ChangeReach}(v) + 0.35 \text{WeightedChangeImpact}(v) + 0.20 \text{NormalizedChangeDepth}(v).$$

- **Relationship (Edge) Removal Oracle ($I_{\text{edge}}(u, v)$)**: the systemic impact of severing one dependency while both endpoints stay operational. Writing $\bar{I}_{\text{comp}}(G)$ for the mean composite impact over G :

$$I_{\text{edge}}(u, v) = \bar{I}_{\text{comp}}(G \setminus \{(u, v)\}) - \bar{I}_{\text{comp}}(G) \quad (13)$$

Topic Criticality Label Masking. FailureSimulator can blend a declared Topic.criticality into its severity term; this is disabled, because that field is a GNN input feature (topic_qos_criticality_ord) and consuming it would score the predictor against a transformation of its own input.

Primary Oracle Declaration and Role Assignment. Because the three reliability-facing oracles (I^* , I_{comp} , I_{dyn}) measure distinct operational constructs, we designate $I^*(v)$ (**FaultInjector**) as the **primary oracle** for all predictive ranking results (Tables 6–8, RQ1–RQ3). $I_{\text{comp}}(v)$ is reserved for Validate-stage quality gates, $I_{\text{dyn}}(v)$ serves as an independent convergent-validity probe, and $I_M(v)$ serves as a structural maintainability reference.

Cross-Oracle Convergent Validity and Critical-Set Bounds. Measured across seven benchmark scenarios and five random seeds on the Application population, the mean Spearman rank correlation is $\rho = 0.907$ for (I_{dyn}, I^*) , $\rho = 0.425$ for (I_{comp}, I^*) , and $\rho = 0.427$ for $(I_{\text{comp}}, I_{\text{dyn}})$. The strong rank agreement ($\rho = 0.907$) between the behavioral queue-flow oracle and the topological cascade injector provides independent convergent evidence across distinct simulation paradigms. However, agreement on the top- K critical set ($K = 0.2n$) is more conservative (mean Jaccard overlap of 0.49 for the strongest pair and 0.24–0.28 for the two I_{comp} pairs, vs. 0.111 expected by chance), highlighting the intrinsic sensitivity of discrete thresholding in non-linear cascades. Consequently, results established against one oracle are never transferred to another; every evaluation metric explicitly references its underlying simulation oracle.

4.4. Input-Label Independence Guarantee

To eliminate data leakage and ensure rigorous evaluation, SaG enforces strict architectural separation between inputs and labels:

- **Feature Space:** Constructed exclusively from G_{analysis} using static structural topology, static code analysis (SCA) metrics, and declared QoS contracts.
- **Label Space:** Evaluated exclusively on raw $G_{\text{structural}}$ through independent simulation oracles (FaultInjector, FailureSimulator, MessageFlowSimulator).

No simulation outputs, failure trace histories, or dynamic execution telemetry are ever exposed as input features to the GNN or the explanation layer.

What this guarantee does and does not establish. The separation rules out circular feature construction: no predictor can read a transformation of the quantity it is scored against. It does not establish statistical independence between features and labels, and we do not claim it does. G_{analysis} is a deterministic projection of $G_{\text{structural}}$ (Section 3.2), so the labels are — up to simulator seed — a deterministic function of the same topology the features are computed from. Two consequences follow, and both bound the results of Section 7.

First, $I^*(v)$ is itself a topological functional: a breadth-first reachability computation over $G_{\text{structural}}$ scaled by a QoS ladder. The predictive task is therefore to recover a closed-form function of a graph from features of that same graph. Read that way, it is unsurprising that an unparameterized centrality score is competitive with a trained model (Section 7.1); the parity we report there is the expected outcome of the setup rather than a surprising failure of graph learning, and we flag it here so the reader does not have to infer it.

Second, and more restrictively, no result in this paper is validated against an observed failure. Every label — on synthetic topologies and on the five open-source systems alike — is simulator-derived. What the evaluation can establish is whether a learned model recovers a simulator’s ordering on architectures it was not trained on. Whether that ordering corresponds to which components actually fail in production is a question this design cannot answer, and Section 8.3 treats it as the study’s principal construct-validity threat.

5. The Explanation Layer: Standards-Grounded Criticality Attribution

The predictor of Section 4 answers *where* to act. It does not answer *what to do*, and neither does the oracle that scores it — both return impact, not cause attributed in standardized quality terms. A component may be critical because it is a single point of failure, because it propagates errors widely, or because it is a

high-coupling maintenance bottleneck. These diagnoses call for distinct architectural repairs: replicating the host or broker, decoupling the topic, or refactoring the module. This section presents the layer supplying that diagnosis. SaG decomposes component and relationship criticality into a standards-grounded quality profile, computed over the same typed node features (Section 3.4) but sharing no parameters with the predictor, and applied to whatever the predictor flagged — triage rather than data flow (Figure 1).

5.1. Grounding in ISO/IEC Standards

In accordance with **ISO/IEC 25010:2023** (Product Quality Model) [9], **ISO/IEC 25019:2023** (Quality-in-Use) [10], and **ISO/IEC 25022:2016** (Measurement of Quality-in-Use) [78], SaG formalizes two primary criticality constructs:

- **Component Criticality (D_1):** The degree to which the sudden failure, unexpected termination, or severe degradation of an individual component reduces the system’s capacity to deliver required services within its operational context of use.
- **Relationship Criticality (D_2):** The degree of systemic service degradation resulting from the severance, partitioning, or failure of a specific dependency or communication channel while both endpoint components remain operational.

Criticality is evaluated across two orthogonal quality characteristics: **Reliability (R)** and **Maintainability (M)**. The two are separated because they imply different repairs, not because they predict different runtime quantities: this layer attributes structural fragility and does not forecast latency, throughput, or queue occupancy. Table 3 outlines this Reliability–Maintainability (RM) quality decomposition, mapping ISO/IEC sub-characteristics to architectural questions, underlying graph metrics, and targeted engineering remediations.

Table 3: The Reliability–Maintainability (RM) quality decomposition.

Dimension	Sub-Characteristic	Architectural Question	Underlying Graph Metrics	Role / Remediation
Reliability (R)	Fault Tolerance (FT)	How broadly does failure propagate?	Reverse PageRank on G^T , in-degree, cascade depth	Reliability Eng.: add redundancy, circuit breakers
	Availability (A)	Is this a single point of failure?	Directed articulation score (raw + QoS-weighted), bridge ratio, connectivity degradation	DevOps/SRE: replicate host/broker
Maintainability (M)	Modularity/Modifiability	How complex and coupled is this?	Betweenness, QoS-weighted out-degree, Code Penalty, coupling-risk imbalance, inverse clustering	Architect: refactor code, decouple

Coverage Scope: SaG focuses specifically on Reliability and Maintainability. Safety (which requires domain-specific hazard logs, such as ISO 26262 Automotive Safety Integrity Level [ASIL] ratings) and Security (which requires explicit threat models, such as STRIDE [Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege]) fall outside purely structural topology analysis and are reserved for domain-specific extensions.

5.2. Composite Quality Score Formulation

All raw topological and code metrics are rank-normalized to $[0, 1]$. Quality sub-characteristics are formulated hierarchically using the Analytic Hierarchy Process (AHP) [53]:

1. **Fault Tolerance ($FT(v)$):** Measures error cascade potential on the transpose graph G_{analysis}^T (where edges follow failure propagation from dependency to dependent):

$$FT(v) = 0.45 \cdot \text{RPR}(v) + 0.30 \cdot \text{Deg}_{\text{in}}(v) + 0.25 \cdot \text{CDPot}_{\text{enh}}(v) \quad (14)$$

where $\text{RPR}(v)$ is Reverse PageRank, $\text{Deg}_{\text{in}}(v)$ is normalized in-degree, and $\text{CDPot}_{\text{enh}}(v)$ is the enhanced Cascade Depth Potential term.

2. **Availability ($A(v)$):** Identifies structural single points of failure (SPOFs) across five terms — directed articulation severity, its QoS-weighted variant, edge-level irrecoverability, connectivity degradation, and the component’s own QoS weight:

$$A(v) = 0.2563 \cdot \text{AP}_c^{\text{dir}}(v) + 0.1998 \cdot \text{QSPOF}(v) + 0.1998 \cdot \text{BR}(v) + 0.2563 \cdot \text{CDI}(v) + 0.0878 \cdot w(v) \quad (15)$$

where $\text{AP}_c^{\text{dir}}(v)$ is Directed Articulation Point severity, $\text{QSPOF}(v)$ is QoS-weighted Single Point of Failure severity, $\text{BR}(v)$ is Bridge Ratio (edge-level irrecoverability), $\text{CDI}(v)$ is Connectivity Degradation Index, and $w(v)$ is the component’s intrinsic QoS weight.

3. **Reliability** ($R(v)$): Blends Fault Tolerance and Availability hierarchically:

$$R(v) = r_\alpha \cdot FT(v) + (1 - r_\alpha) \cdot A(v), \quad r_\alpha = 0.36 \quad (16)$$

The blend weight $r_\alpha = 0.36$ is screened in Supplementary Section S1. The intra-dimension weights apply $\lambda = 0.70$ shrinkage blending with a uniform prior (Section 7.3). Because three of the comparison matrices are rank-one by construction (implying near-zero CR by construction, detailed in Supplementary Section S4), these weights represent declared constants with documented internal structure rather than independently elicited expert judgments.

4. **Maintainability** ($M(v)$): Evaluates structural coupling combined with code-level static analysis across five terms:

$$M(v) = 0.35 \cdot BT(v) + 0.30 \cdot w_{\text{out}}(v) + 0.15 \cdot CQP(v) + 0.12 \cdot \text{CouplingRisk}_{\text{enh}}(v) + 0.08 \cdot (1 - CC(v)) \quad (17)$$

where $BT(v)$ is Betweenness Centrality, $w_{\text{out}}(v)$ is QoS-weighted efferent coupling, $CQP(v)$ is Code Quality Penalty, $\text{CouplingRisk}_{\text{enh}}(v)$ is coupling imbalance, and $CC(v)$ is local Clustering Coefficient.

The baseline composite quality score $Q(v)$ combines both dimensions:

$$Q(v) = 0.80 \cdot R(v) + 0.20 \cdot M(v) \quad (18)$$

When evaluating under a specific ISO/IEC 25019 Context of Use vector $\vec{\omega} = [q_R, q_M]^\top$, the score is reweighted dynamically:

$$Q_{\text{domain}}(v) = q_R \cdot R(v) + q_M \cdot M_{\text{static}}(v) \quad (19)$$

Components are categorized into adaptive criticality tiers using Tukey quartile thresholds: **CRITICAL** ($Q > Q_3 + 1.5 \cdot \text{IQR}$), **HIGH** ($Q_3 < Q \leq Q_3 + 1.5 \cdot \text{IQR}$), **MEDIUM** ($Q_1 < Q \leq Q_3$), and **MINIMAL** ($Q \leq Q_1$). This yields an actionable distinction: high A with low FT indicates a single point of failure calling for horizontal replication, whereas high FT denotes an error-cascade hub requiring circuit breakers, rate limiting, and bulkhead isolation (Section 8.4).

6. Experimental Setup

6.1. Datasets and System Corpus

The evaluation corpus comprises 2,812 components across seventeen system architectures — twelve synthetic topologies that form the inductive cross-validation folds, and five real-world reference systems withheld entirely from training — as detailed in Table 4:

Here, $|V|$ is the sum of all five entity-type counts per scenario. The twelve synthetic topologies total 2,461 components; the eleven carrying the manifest’s **evaluation** role account for 2,387 of these and the ATM case study for the remaining 74. The five real-world systems add 351. $|E|$ counts every raw structural relationship instance recorded in the scenario specification — the native substrate that simulation oracles traverse — rather than the derived **DEPENDS_ON** projection constructed for GNN training. All six structural relation types are included in that count (**PUBLISHES_TO**, **SUBSCRIBES_TO**, **ROUTES**, **RUNS_ON**, **CONNECTS_TO**, **USES**); we note the convention explicitly because the generator’s own build log omits **CONNECTS_TO** and therefore reports slightly lower per-scenario totals. Every count in Table 4 is read directly from the committed topology files, whose byte-identical regeneration from configuration is asserted in continuous integration (Section 6.1.1).

Five real-world architectures were transcribed from authentic open-source repositories using dedicated architectural adapters. The synthetic scenarios were produced by a parameterized topology generator: each is fully defined by a committed configuration specifying a random seed, per-entity-type counts, seven-number summaries (mean, median, standard deviation, minimum, maximum, Q_1 , Q_3) for application publish and subscribe fan-out, applications per host, library fan-in, topic payload size, and categorical distributions over the three QoS dimensions. Graph degree distributions and clustering emerge directly from these parameters rather than being synthetically forced. Supplementary Section S5 reports the generative parameters governing each topology’s shape; complete configurations are included in the replication package.

Corpus Composition across Experimental Regimes. The experimental evaluation spans three complementary regimes with precisely bounded corpora:

Table 4: Experimental evaluation corpus. The twelve synthetic topologies are the inductive Leave-One-Scenario-Out folds of Table 8; the five real-world systems are withheld from every training fold and used only for zero-shot transfer (Section 7.4). Counts are read from the committed topology files rather than from the generator configurations.

Dataset / Architecture	System Paradigm	$ V $	$ V_{app} $	Topics	Brokers	Hosts	Libs	$ E $
<i>Synthetic evaluation scenarios (evaluation role in the manifest)</i>								
Autonomous Vehicle (AV)	ROS 2 Cyber-Physical	152	80	40	4	8	20	773
Enterprise Pub-Sub	Kafka Event Mesh	520	300	120	10	40	50	3,389
Financial Trading	Low-Latency Pub-Sub	124	60	35	5	6	18	583
Healthcare Integration	HL7/FHIR Event Mesh	98	50	25	3	8	12	428
Hub-and-Spoke Enterprise	Broker-Centric Messaging	139	70	30	2	12	25	751
IoT Smart City	MQTT Telemetry Mesh	326	200	80	6	30	10	1,331
Microservices Mesh	Cloud-Native Services	186	90	45	6	15	30	680
Telecom RAN	5G Radio Access Network	225	120	55	8	20	22	858
Industrial SCADA	Plant Control Telemetry	254	140	70	4	25	15	818
Real-Time Gaming	Multiplayer State Sync	158	75	38	5	12	28	643
Logistics Fleet	Vehicle Telematics Mesh	205	110	50	7	18	20	792
<i>Synthetic case study (LOSO fold; case_study role in the manifest)</i>								
Air Traffic Management (ATM)	ICAO Global ATM Concept	74	26	27	5	8	8	271
<i>Real-world reference systems (never used as training folds)</i>								
Autoware.universe [79]	Real-World ROS 2 Autoware	75	32	24	3	6	10	179
Cloud Microservices [80]	Real-World GCP Boutique	60	22	20	4	6	8	128
Train-Ticket [81]	Real-World Microservices	90	41	30	3	8	8	162
Home Assistant [82]	Real-World Smart Home	63	24	22	3	6	8	119
EdgeX Foundry [83]	Real-World Industrial IoT	63	22	24	3	6	8	112
Synthetic subtotal (12 LOSO folds)		2,461	1,321	615	65	202	258	11,317
Real-world subtotal (5 systems)		351	141	120	16	32	42	700
Total		2,812						12,017

1. *In-Distribution Evaluation (Table 6)*: Evaluated on the seven core synthetic domains from Table 4 using stratified 60% train / 20% validation / 20% test node splits over five random seeds.
2. *Inductive Leave-One-Scenario-Out (LOSO) Cross-Validation (Table 8)*: Evaluated across twelve distinct inductive folds totaling 2,461 components: the seven core synthetic scenarios, four extended domain topologies (Telecom RAN, Industrial SCADA, Realtime Gaming, and Logistics Fleet) — 2,387 components between them — and an Air Traffic Management (ATM) network scenario contributing the remaining 74. In each fold, models are trained on eleven graphs and tested zero-shot on the held-out twelfth graph.
3. *Real-World Architectural Transfer (Table 11; Supplementary Section S7)*: The five open-source real-world systems (Autoware.universe, Cloud Microservices, Train-Ticket, Home Assistant, EdgeX Foundry) are never used as training folds; they are withheld entirely and used strictly for zero-shot architectural transfer validation.

Not every analysis in Section 7 runs on the full corpus: the sensitivity sweeps and the detection benchmark predate the four extended domains and operate on smaller cached subsets. Because a reader comparing figures across subsections would otherwise have no way to tell which population a number belongs to, Table 5 states the subset used by each analysis. Comparisons are made only within a row.

6.1.1. Reproducibility of the Corpus

The benchmark corpus is designed to be fully regenerable rather than merely statically archived. Each dataset is deterministically generated from its configuration file via:

```
python cli/generate_graph.py batch -input-dir data/scenarios -output-dir <dir>
```

A companion manifest records the random seed, entity counts, git commit hash, and a SHA-256 cryptographic digest for each emitted topology. Continuous integration regression tests assert that every committed dataset regenerates *byte-identically* from its configuration and that all disk digests match the manifest. This guarantees that third parties can reproduce the exact graphs used in our experiments, rather than simply sampling from similar distributions.

Table 5: Which corpus subset backs each analysis. Figures from different rows are not directly comparable, and we do not compare them.

Analysis	Scenario subset	n	Reported in
In-distribution ranking	Seven core synthetic domains	7	Tables 6–7
Inductive LOSO	Eleven evaluation scenarios + ATM	12	Table 8, Sections 7.1–7.2
QoS edge-feature ablation	Same twelve LOSO folds	12	Section 7.3.1
Weight sweeps and Morris screening	Six–seven core synthetic domains	6–7	Supplementary S1
Cross-oracle convergent validity	Seven core synthetic domains	7	Table 10
Anti-pattern detection, stratification	Seven core domains + ATM	8	Section 7.3.3
Relational attention illustration	ATM case study alone	1	Supplementary S8
Real-world zero-shot transfer	Five open-source systems	5	Table 11, Supp. S7

6.2. Baselines and Evaluated Predictors

We evaluate four primary predictor configurations drawn from three families. Predictor names state the family and the substrate: an $-N$ suffix marks a model trained on the *native* multigraph, its absence the derived Application–Library flow projection, and a $-QoS$ suffix marks a configuration that consumes declared QoS contracts. *SaG* throughout denotes the framework, never an individual predictor.

- Heterogeneous graph learning (typed HGT). HGT-QoS** (proposed): relation-specific Heterogeneous Graph Transformer (Section 4) ingesting the complete native multigraph with 16-dimensional continuous-categorical edge features that encode middleware QoS contracts. Its ablation **HGT**, which masks those QoS dimensions, is reported in Section 7.3.1.
- Homogeneous graph learning (untyped GAT). GAT-N-QoS**: homogeneous Graph Attention Network [62] trained on the identical native multigraph substrate with per-type input projections, but untyped, single-relation message passing. Its edge channel carries the scalar QoS aggregate $w(e)$ — dimension 0 of the same 16-D encoding HGT-QoS consumes — rather than the per-dimension decomposition; no homogeneous architecture in our suite ingests the full 16-D vector. The HGT-QoS–GAT-N-QoS contrast therefore bounds the *joint* contribution of relational typing and per-dimension QoS encoding. Section 7.3.1 separates the second factor within the typed architecture, and the corresponding unweighted ablation is **GAT-N**.
- QoS-weighted structural baseline (training-free). Topo-QoS**: QoS-weighted topological centrality evaluated on the derived application flow projection.
- Unweighted structural baseline (training-free). Topo**: structural centrality combining unweighted betweenness centrality and articulation point scoring on the flow projection.

In addition, the out-of-distribution evaluation (Table 8) reports **RM** ($Q(v)$, the deterministic hierarchical quality attribution model of Section 5) as a diagnostic reference baseline. RM is not fitted to rank failure impact; its inclusion demonstrates how much learned relational prediction adds over static structural attribution (Section 1.2). Furthermore, deterministic RM scoring drives every sensitivity sweep in Section 7.3, where closed-form formulations isolate parameter effects from neural training stochasticity.

6.2.1. Evaluation Substrates and Parity Guarantee

Predictors in this study operate on matched substrates within their respective families:

- Graph Learning Models (GAT-N-QoS, HGT-QoS)**: Both learned neural predictors ingest the complete native typed multigraph across all five entity types in both in-distribution (Table 6) and out-of-distribution Leave-One-Scenario-Out (Table 8) evaluations — which is what the shared $-N$ suffix records — so the comparison carries no multi-entity visibility confound. Node type reaches homogeneous GAT-N-QoS only through its per-type input projection layer, while message passing uses untyped GATConv with shared weights across all edges; in contrast, heterogeneous HGT-QoS employs relation-specific HGTCov weight matrices per edge triple alongside edge-type encodings. Substrate and node features are matched; the edge channel is not, since GAT-N-QoS consumes the scalar QoS

aggregate $w(e)$ where HGT-QoS consumes all 16 dimensions (Section 6.2). Comparisons between the two therefore isolate relation-specific parameterization jointly with per-dimension QoS encoding, and we report the QoS factor separately in Section 7.3.1 rather than attributing the whole margin to typing.

- **Training-Free Structural Baselines (Topo, Topo-QoS):** Topological baselines are evaluated on the derived Application–Library `DEPENDS_ON` projection (Section 3.2). This projected substrate is necessary because in raw publish–subscribe multigraphs, Application nodes never route messages directly, resulting in near-zero betweenness and bridge ratios that yield degenerate, uninformative scores.
- **Ground-Truth Independence Guarantee:** Crucially, **no predictor in either group accesses $G_{\text{structural}}$** : that raw topology is strictly reserved as the substrate for ground-truth simulation oracles (Section 4.4), a guarantee formally verified by `tests/test_independence_guarantee.py`.

Regardless of substrate, all variants are scored on an identical, independently resolved Application node set (V_{app} , Section 6.3).

6.3. Evaluation Metrics and Protocols

Figure accessibility. All figures in this paper use the high-contrast, colorblind-safe Okabe–Ito palette together with distinct marker, hatching, and node-shape encodings, so that every distinction carried by color is also carried by form and remains legible in monochrome.

- **Ranking Precision:** Evaluated via Spearman rank correlation (ρ) and Kendall’s rank correlation (τ) between predicted component rankings and ground-truth simulated impact $I^*(v)$ from the primary oracle (Section 4.3).
- **Critical-Set Identification:** Measured via $F_1@K$, Precision@ K , and Recall@ K for top- K critical components, where $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$. Because predicted and ground-truth sets both contain exactly K elements, Precision, Recall, and F_1 coincide identically as the top- K set overlap.
- **Statistical Significance:** Assessed through paired Wilcoxon signed-rank tests [84] ($p < 0.05$) and non-parametric bootstrap 95% confidence intervals ($B = 2,000$) over folds [85, 86]. In the 12-fold LOSO design, power floor is $p = 0.00049$. Applying Holm’s step-down correction across the ten full-population rank contrasts in Sections 7.1–7.3.1, three survive: Topo-QoS over Topo ($p = 0.0010$), Topo over RM ($p = 0.0005$), and unweighted typing HGT over GAT-N ($p = 0.0010$), while QoS-weighted typing ($p = 0.0122$) and QoS edge ablation ($p = 0.0093$) remain nominally significant with 11/12 directional fold consistency.

Pre-registration. The primary out-of-distribution contrast (HGT-QoS vs. Topo-QoS under LOSO, 5 fixed seeds, fold as unit of analysis) was pre-registered before results were obtained, committing to report measured outcomes regardless of significance. As reported in Section 7.1, the margin did not clear statistical significance.

6.3.1. Evaluation Population

Every predictor within a given evaluation table is scored on an identical node population, resolved strictly from scenario topology and simulation ground truth — never from any model’s predictions. Unless otherwise noted, this population is the **Application** set (V_{app}). This aligns with the framework’s primary objective (forecasting application-layer cascading failures) and ensures a fair common denominator across both typed and untyped predictors. Pooling node types into a single global ranking conflates distinct base rates and impact distributions, shifting the resulting rank correlation outside the envelope of per-type correlations (Section 7.3). We therefore report stratified, single-population metrics throughout and explicitly identify any pooled figures.

6.3.2. Evaluation Protocols

- **In-Distribution Evaluation:** 60% train / 20% validation / 20% test node splits over five seeds {42, 123, 456, 789, 2024}. Splits are deterministic functions of node ID and seed, inducing partition and training noise reflected in reported standard deviations.
- **Inductive Leave-One-Scenario-Out (LOSO):** Models train on eleven scenarios and test zero-shot on the held-out twelfth across all 12 folds. Parity is strictly enforced: every learned variant receives identical training sets of $N - 1$ graphs, message-passing depth is fixed at three layers, and checkpoint selection uses an inner validation split within the primary graph under an identical early-stopping protocol (Section 8.4). The outer holdout scenario participates in no training or selection decisions.
- **Real-World Architectural Transfer:** Models trained on synthetic topologies evaluate zero-shot on five open-source distributed systems without fine-tuning.

7. Results and Empirical Analysis

This section presents empirical results for RQ1–RQ5 across the twelve-fold inductive benchmark and five authentic open-source distributed systems. Evaluated populations are strictly stratified on the Application service set (V_{app}) under the input–label independence guarantee (Section 4.4).

7.1. RQ1: Graph Learning vs. Structural Baselines

Table 6 presents in-distribution held-out Spearman rank correlation (ρ) against simulated cascade impact $I^*(v)$ across seven representative distributed architecture domains.

Table 6: In-distribution held-out ρ against $I^*(v)$: mean over five seeds $\pm$ spread; n = held-out Application count. All four learned variants use the identical native multigraph, differing only in typing and edge channel (Section 6.2); topological baselines use the flow projection. Each seed redraws the 60/20/20 split as well as the initialization. Read with Table 7.

Scenario	n	Topo	Topo-QoS	GAT-N	GAT-N-QoS	HGT	HGT-QoS
AV System	16	0.362 $\pm$ 0.169	0.762 $\pm$ 0.159	0.632 $\pm$ 0.224	0.758 $\pm$ 0.110	0.797 $\pm$ 0.092	0.737 $\pm$ 0.126
Enterprise	60	0.421 $\pm$ 0.089	0.803 $\pm$ 0.060	0.638 $\pm$ 0.355	0.864 $\pm$ 0.033	0.726 $\pm$ 0.381	0.903 $\pm$ 0.014
Financial Trading	12	0.169 $\pm$ 0.375	0.636 $\pm$ 0.125	0.864 $\pm$ 0.047	0.573 $\pm$ 0.483	0.758 $\pm$ 0.189	0.590 $\pm$ 0.245
Healthcare	10	−0.275 $\pm$ 0.267	0.660 $\pm$ 0.190	0.863 $\pm$ 0.039	0.802 $\pm$ 0.068	0.475 $\pm$ 0.387	0.478 $\pm$ 0.327
Hub-and-Spoke	14	0.025 $\pm$ 0.230	0.545 $\pm$ 0.260	0.538 $\pm$ 0.219	0.441 $\pm$ 0.308	0.367 $\pm$ 0.571	0.425 $\pm$ 0.462
IoT Smart City	40	0.124 $\pm$ 0.129	0.361 $\pm$ 0.082	0.790 $\pm$ 0.103	0.639 $\pm$ 0.191	0.825 $\pm$ 0.053	0.848 $\pm$ 0.045
Microservices	18	0.337 $\pm$ 0.202	0.635 $\pm$ 0.136	0.514 $\pm$ 0.105	0.497 $\pm$ 0.209	0.421 $\pm$ 0.301	0.428 $\pm$ 0.327
Mean	—	0.166	0.629	0.691	0.653	0.624	0.630

Table 7: Paired Wilcoxon signed-rank tests across in-distribution scenarios ($n = 7$, two-sided).

Comparison	$\Delta\rho$	Won	Wilcoxon W	p -value	Significance
HGT-QoS vs. Topo	+0.464	7/7	0.0	0.0156	Significant ($p < 0.05$)
Topo-QoS vs. Topo	+0.463	7/7	0.0	0.0156	Significant ($p < 0.05$)
GAT-N vs. HGT-QoS	+0.061	4/7	9.0	0.469	Not significant
HGT-QoS vs. GAT-N-QoS	−0.023	3/7	12.0	0.813	Not significant
HGT vs. GAT-N	−0.067	3/7	8.0	0.375	Not significant
HGT-QoS vs. HGT	+0.006	5/7	11.0	0.688	Not significant
HGT-QoS vs. Topo-QoS	+0.001	2/7	10.0	0.578	Not significant
GAT-N-QoS vs. Topo-QoS	+0.024	3/7	13.0	0.938	Not significant

Table 8: Inductive LOSO evaluation, Application population, twelve folds. Learned variants share training set, substrate, depth, and selection rule (Section 6.3), differing only in typing and edge channel. Fold score = mean over five seeds; **Fold** σ = spread of the twelve fold means; **Seed** σ = median within-fold spread (zero by construction for deterministic baselines); CI = bootstrap over folds ($B = 2,000$).

Predictor / Reference	Mean LOSO ρ	95% CI	Fold σ	Seed σ	Critical-Set $F_1@K$	Requires Training
<i>Training-free structural baselines</i>						
Topo	0.250	[0.128, 0.356]	0.200	—	0.306	No
Topo-QoS	0.568	[0.418, 0.686]	0.237	—	0.353	No
<i>Learned predictors (shared native substrate, matched training set)</i>						
GAT-N	0.493	[0.437, 0.541]	0.092	0.158	0.417	Yes
GAT-N-QoS	0.581	[0.493, 0.657]	0.141	0.017	0.474	Yes
HGT	0.640	[0.565, 0.716]	0.138	0.069	0.466	Yes
HGT-QoS	0.695	[0.631, 0.748]	0.103	0.053	0.507	Yes
<i>Diagnostic reference — not a ranking model</i>						
RM / $Q(v)$	0.133	[0.009, 0.247]	0.215	—	0.258	No

7.1.1. Out-of-Distribution (LOSO) Generalization

In inductive Leave-One-Scenario-Out (LOSO) cross-validation, models are evaluated on their capacity to predict cascading criticality over completely unseen system topologies:

Twelve LOSO folds are reported, comprising the eleven synthetic evaluation scenarios and the ATM case study, all specified in Table 4 (Section 6.1). In each fold, one scenario is held out for zero-shot testing while the model is trained exclusively on the remaining eleven. All variants are evaluated on the identical Application node set per fold (Section 6.3); paired Wilcoxon tests are conducted across the twelve folds, where the smallest attainable two-sided p is 0.00049. Per-fold evaluated populations range from 26 to 300 Application nodes, so $K = \text{round}(0.20 |V_{\text{app}}|)$ ranges from 5 to 60. On $F_1@K$, HGT-QoS beats Topo-QoS in 8 of 12 folds ($\Delta = +0.154$, $W = 12.0$, $p = 0.034$) but separates from untyped GAT-N-QoS in 8 of 12 without reaching significance ($\Delta = +0.034$, $W = 25.0$, $p = 0.301$): critical-set identification distinguishes the typed learned model from the training-free baseline, but not from untyped learning.

Label-noise ceiling. These correlations are bounded by the reproducibility of the target they are scored against. Re-running the ground-truth oracle across the five seeds gives a test-retest rank correlation between 0.880 and 1.000 across the twelve folds (median 0.979; ten of twelve at or above 0.95), with Microservices the least reproducible at 0.880. HGT-QoS’s $\rho = 0.695$ therefore recovers roughly 71% of the attainable signal against the median ceiling, and no predictor in Table 8 can exceed the reproducibility of its own labels. Top- K critical sets are the noisier construct by a wide margin: their cross-seed Jaccard has a median of 0.709 and falls to 0.400 (Microservices), 0.467 (Telecom RAN), and 0.481 (Logistics Fleet). That instability is the main reason the $F_1@K$ margins are less stable than the ranking margins, and it bounds how much weight any single critical-set comparison can carry. Notably, Microservices is both the least reproducible fold and one of the two on which typed learning loses (Section 7.2.1) — part of that deficit may be label noise rather than model failure.

Figure 3 summarizes these results alongside critical-set identification and inter-oracle agreement.

Key Insights for RQ1:

- Typed learning is the best configuration, but not demonstrably better than the QoS baseline.** HGT-QoS leads all predictors out-of-distribution ($\rho = 0.695$), and the typed-vs-untyped margin excludes zero decisively (Section 7.2). Against training-free *Topo-QoS*, HGT-QoS is $+0.127$ (9/12, $W = 16.0$, $p = 0.077$, CI $[+0.011, +0.255]$) and HGT is $+0.073$ (10/12, $p = 0.129$, CI $[-0.038, +0.196]$). We decline to read the first as a win: the margin is carried almost entirely by ATM, where Topo-QoS fails outright ($\rho = -0.086$, its only negative fold) against HGT-QoS’s 0.579. Excluding that fold the margin falls to $+0.078$ (8/11, $p = 0.148$). A bootstrap interval that excludes zero on the strength of one fold is not evidence of general superiority, and we report interval and sensitivity together rather than quoting whichever is more favorable.
- A QoS-weighted structural score is a genuinely strong baseline.** Topo-QoS reaches $\rho = 0.568$ zero-shot, beating unweighted Topo on 11 of 12 folds ($+0.318$, $p = 0.0010$) and remaining

indistinguishable from untyped learning in both directions (GAT-N trails by 0.075, $p = 0.077$; GAT-N-QoS leads by 0.013, $p = 0.850$). Any claim that graph learning is *required* must be made against this baseline, not against unweighted centrality.

3. **Power is not the limiting factor.** At $n = 12$ the design tolerates four lost folds and still reaches $\alpha = 0.05$, provided the losses are smallest in magnitude. HGT-QoS's are not: it loses Enterprise (-0.268), Microservices (-0.080), and Real-Time Gaming (-0.004), and Enterprise is the largest $|\Delta|$ among them — which is what holds W at 16.0. Enlarging the corpus will not resolve this; the two substantive inversions must be understood instead (Section 7.2.1).
4. **The explanation layer is weakly predictive, not noise.** $RM/Q(v)$ reaches $\rho = 0.133$, losing to unweighted Topo on all twelve folds (-0.117 , $p = 0.0005$), so no ranking claim is made for it. Its interval $[0.009, 0.247]$ stays above zero and it supplies interpretable diagnostics without training (Section 5). It appears in Table 8 as a reference point, not a competitor.

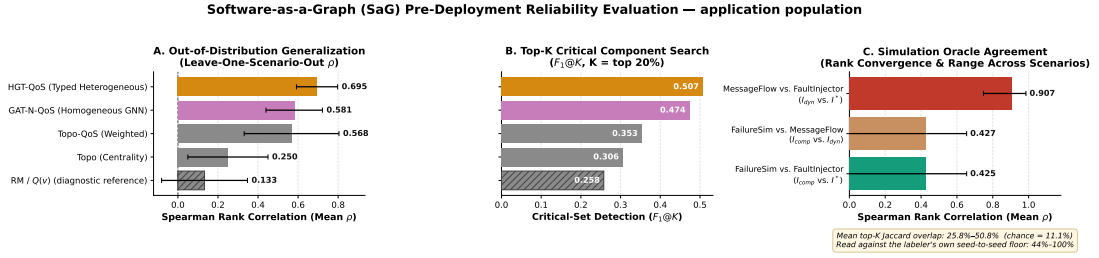


Figure 3: Results at a glance, Application population. (A) Out-of-distribution rank correlation per predictor across the twelve LOSO folds. (B) Critical-set identification at $K = 20\%$. (C) Pairwise rank agreement between the three simulation oracles, against the chance baseline. Panels A and B are read directly from the same artifact as Table 8 and panel C from that behind Table 10; the ordering shown is whatever the data gives, which is why the untrained Topo-QoS baseline sits third rather than last.

7.1.2. The Active Stratum: Ranking Quality Separated From Inertness Detection

Every LOSO figure above is a correlation over the full held-out Application population, and between 21% (Microservices) and 52% (Healthcare) of that population carries exactly zero simulated impact depending on the fold. A predictor can therefore score well by separating components that can propagate a failure from those that cannot, without ordering the propagating ones correctly. Because these are different capabilities with different operational value, we re-score all twelve folds on the *active stratum* — the $n_{>0}$ components with strictly positive ground-truth impact — using the same predictions, folds, and seeds. Table 9 reports both.

Table 9: LOSO means over twelve folds on the full Application population (ρ) and restricted to components with strictly positive ground-truth impact ($\rho_{>0}$). Same predictions, folds, and seeds; only the evaluated subset differs. Training-free baselines lose most of their apparent accuracy under the restriction; learned models lose far less.

Predictor	ρ (full)	$\rho_{>0}$ (active)	Retained
RM / Q(v)	0.133	0.014	11%
Topo	0.250	0.064	26%
Topo-QoS	0.568	0.183	32%
GAT-N	0.493	0.322	65%
GAT-N-QoS	0.581	0.338	58%
HGT	0.640	0.382	60%
HGT-QoS	0.695	0.407	59%

Three consequences follow, and they do not all favor the proposed model.

1. **The training-free baselines are substantially weaker than their headline figures suggest.** Topo-QoS retains only 32% of its full-population correlation on the active stratum, against 59% for

HGT-QoS, and it turns negative on three folds (ATM -0.556 , Healthcare -0.248 , and Topo on Financial Trading -0.185). Much of what a QoS-weighted centrality score contributes is the detection of inert components, which the degree-zero structure of the projection makes nearly free. This is the clearest evidence in the paper that learning contributes something a structural heuristic does not.

2. **RQ1’s verdict is nonetheless unchanged.** On the active stratum HGT-QoS leads Topo-QoS by $+0.224$ (9/12, $W = 18.0$, $p = 0.110$) — a margin nearly twice the full-population $+0.127$, but with enough fold-level variance that it still does not reach significance. We report it as a widened but statistically undemonstrated margin, and the RQ1 conclusion of Section 7.1 stands as written.
3. **The typed-versus-untyped result survives; the QoS ablation does not.** Typing remains significant under the restriction (HGT-QoS vs. GAT-N-QoS $+0.069$, 10/12, $p = 0.043$; HGT vs. GAT-N $+0.060$, 10/12, $p = 0.027$), so the central claim of Section 7.2 is not an artifact of zero-inflation. The QoS *encoding*, by contrast, loses its significance entirely: the $+0.054$ typed gain ($p = 0.009$) falls to $+0.025$ ($p = 0.151$), and the untyped gain from $+0.088$ ($p = 0.034$) to $+0.017$ ($p = 0.791$). We therefore narrow the RQ3 ablation claim in Section 7.3: declared QoS attributes help identify *which* components can propagate a failure, and we have no evidence that they help rank the ones that do.

7.2. RQ2: Value of Typed Heterogeneity

To evaluate the specific contribution of node and edge typing, we contrast the relation-specific Heterogeneous Graph Transformer against the homogeneous Graph Attention Network on the shared native multigraph substrate, with the identical training set, depth, and model-selection rule:

- **In-Distribution Fitting (Table 6): typing does not help.** On familiar architectures, typed message passing carries no advantage at all. HGT-QoS reaches $\rho = 0.630$ against GAT-N-QoS’s 0.653 ($\Delta\rho = -0.023$, won in 3 of 7, $W = 12.0$, $p = 0.813$), and the unweighted pair runs the same way (HGT 0.624 vs. GAT-N 0.691 , -0.067 , $p = 0.375$). The best in-distribution mean in the table belongs to *untyped* GAT-N. None of these differences is significant at $n = 7$, and per-seed dispersion is large (median within-scenario σ of 0.105 – 0.301 across the learned variants, reaching 0.571 on Hub-and-Spoke), so the honest reading is parity rather than a reversal — but there is no in-distribution typing benefit to report.
- **Out-of-Distribution Generalization (LOSO, Table 8):** Under inductive distribution shift, the typed advantage is the most robust effect in this study. HGT-QoS outperforms GAT-N-QoS by $+0.114$ ($\rho = 0.695$ vs. 0.581), winning 11 of 12 folds ($W = 8.0$, $p = 0.0122$; 95% CI $[+0.048, +0.170]$). The unweighted pair replicates it and slightly exceeds it: HGT over GAT-N by $+0.147$, 11 of 12 folds, $W = 1.0$, $p = 0.0010$, CI $[+0.101, +0.185]$.

Typing is an inductive bias, not a capacity advantage. The contrast between the two regimes is the finding. Given training data from the same generator as the test split, untyped message passing matches typed and nominally exceeds it. Asked to transfer to an unseen architecture, the typed model wins 11 folds of 12 by $+0.114$, while the untyped model loses 0.088 of its in-distribution standing ($0.653 \rightarrow 0.581$) and the typed model gains 0.065 ($0.630 \rightarrow 0.695$). The reading is that relation-specific parameters add no fitting capacity — with enough same-distribution data an untyped attention mechanism recovers the same structure — but encode which distinctions survive a change of topology. Distinguishing PUBLISHES_TO dissemination from RUNS_ON placement is a constraint rather than extra expressiveness, and constraints pay off exactly where distribution shift would otherwise mislead. This is a more specific claim than we set out to test, and a more useful one for a pre-deployment gate, where every architecture analyzed is by construction new.

The single fold typed learning loses is the same in both contrasts: ATM, the smallest graph in the corpus (26 Applications) and the only one designated a case study. Excluding it, both contrasts win every remaining fold ($+0.139$ and $+0.161$, 11/11, $p = 0.0010$). We report the twelve-fold figure as the headline regardless, because excluding an inconvenient fold to recover unanimity is precisely the move a reader should distrust.

Methodological Controls and Substrate Parity. To ensure that the typed–untyped margin reflects genuine architectural inductive biases rather than experimental artifacts, all learned predictors in Table 8 operate under strict substrate and training-set parity: every model receives all $N - 1$ training graphs, message-passing depth is fixed at three layers, and checkpoint selection follows the same rule — a validation split within the primary training graph (Section 6.3) — for every variant alike. Under these controlled conditions, homogeneous GAT-N reaches $\rho = 0.493$, so the observed heterogeneous advantage ($\Delta\rho = +0.114$, won in 11 of 12 folds, $p = 0.0122$) is attributable to relational typing rather than to substrate, training set, depth, or selection rule.

7.2.1. Where Typed Learning Fails, and How It Can Be Detected

HGT-QoS loses to Topo-QoS on three folds, but only two of them are substantive: Enterprise ($\rho = 0.569$ vs. 0.838) and Microservices (0.483 vs. 0.563). The third, Real-Time Gaming (0.776 vs. 0.780), is a tie to within 0.004 and carries no interpretive weight. It is the two substantive inversions that hold the RQ1 comparison below significance, and both are folds on which the training-free baseline is unusually strong, which suggests the model is discarding structural signal the baseline retains rather than failing to learn.

Absence of a Label-Free Confidence Signature. We evaluated whether the standard deviation of predicted scores $\hat{\sigma}$ over held-out applications could signal model reliability at inference time without labels. Although the two worst-performing folds (Enterprise, $\hat{\sigma} = 0.111$; Microservices, 0.138) exhibit low dispersion, the correlation does not hold across the full benchmark: $\hat{\sigma}$ correlates with the margin over Topo-QoS at $\rho_s = -0.126$ ($p = 0.697$) for HGT-QoS, and the second-lowest dispersion fold (Healthcare, 0.123) yields one of the largest positive margins (+0.284). Neither graph size (rank correlation with margin -0.357 , $p = 0.255$) nor edge density reliably flags fold difficulty in advance.

Feature scale drift across scenarios (documented in `results/feature_shift_diagnostic.md`) remains the primary explanation for the Enterprise deficit, as Enterprise is the largest graph (520 nodes) where feature scaling disparities are most acute. Because model hyperparameters are strictly held constant across folds by protocol, a practitioner currently has no automated label-free signal to pre-determine whether an unseen architecture will favor the learned model or the training-free baseline (Section 8.4).

7.3. RQ3: Ablations and Sensitivity Analysis

This section reports the ablations that bear on a headline claim — the QoS edge encoding, cross-oracle agreement, and the per-type stratification that governs how every other result in this paper is read. The parameter-sensitivity sweeps over the explanation layer’s ten declared weight constants establish robustness rather than any finding of their own, and are reported in full in the supplementary material (Supplementary Sections S1–S2). Their collective result is stated here so the body remains self-contained: of the ten constants, only the Fault-Tolerance/Availability blend r_α and the AHP shrinkage λ carry appreciable influence on ρ ($\mu^* = 0.144$ and 0.117 under Morris screening, against ≤ 0.023 for the remaining eight), and no setting of the topic-weight or QoS sub-weight constants would change any comparison reported above. The elicited AHP weights are, notably, *anti*-predictive: rank correlation falls monotonically from 0.262 under a uniform prior to 0.166 under raw AHP judgment. We retain them because RM is an attribution instrument rather than a ranking model, and discuss that trade in Section 8.4.

7.3.1. QoS Feature Ablation

To isolate the specific empirical contribution of the continuous-categorical QoS edge features (Section 4.1.1), we evaluated **HGT**, an un-augmented ablation of HGT-QoS whose edge features contain only scalar coupling and relation one-hot encodings.

Under the inductive LOSO evaluation, the QoS edge encodings carry a ranking benefit in both architectures on the full held-out population. Mean LOSO rank correlation is $\rho = 0.695$ with the encodings and 0.640 without them ($\Delta\rho = +0.054$, won in 11 of 12 folds, $W = 7.0$, $p = 0.0093$; 95% bootstrap CI [+0.021, +0.091]). The homogeneous architecture agrees in direction and magnitude: GAT-N-QoS achieves 0.581 against GAT-N 0.493 ($\Delta\rho = +0.088$, won in 9 of 12 folds, $W = 12.0$, $p = 0.034$, CI [+0.021, +0.151]). The typed result is robust to dropping the ATM fold (+0.049, 10 of 11, $p = 0.019$); the homogeneous one is not (+0.071, 8 of 11, $p = 0.067$), so we treat the typed evidence as the finding and the homogeneous replication as supporting.

The gain does not survive restriction to the active stratum, and we do not claim it does. Re-scoring the identical folds on the $n_{>0}$ components that carry strictly positive impact (Section 7.1.2) removes the effect in both architectures: the typed gain falls from +0.054 ($p = 0.0093$) to +0.025 (8 of 12, $W = 20.0$, $p = 0.151$), and the homogeneous gain from +0.088 ($p = 0.034$) to +0.017 (6 of 12, $W = 35.0$, $p = 0.791$) — the latter no better than a coin flip across folds. The full-population effect is therefore real but narrower than it first appears: declared QoS attributes help the model decide *which* components can propagate a failure at all, and this evaluation provides no evidence that they help order the components that do. That is a weaker claim than the one an earlier version of this section made, and it is the claim the data support. It matters for deployment, because a practitioner reading Table 8 would otherwise expect the QoS channel to sharpen a criticality ranking, which is the use it cannot be shown to serve. The typed-versus-untyped result, by contrast, does survive the same restriction (Section 7.1.2), which is why we treat typing rather than QoS encoding as this paper’s load-bearing architectural claim.

The encodings also improve optimization reproducibility, though less symmetrically than the ranking result. The median within-fold standard deviation across five random seeds is 0.053 for HGT-QoS against 0.069 for un-augmented HGT, and 0.017 for GAT-N-QoS against 0.158 for GAT-N — a large effect in the homogeneous pair and a modest one in the typed pair. We read the ranking gain as the primary result and the variance reduction as a secondary property, rather than the reverse.

QoS Parameter Variance. Modal QoS shares range from 29% to 89% across the twelve scenarios (Supplementary Section S5), ensuring that every fold carries genuine variation in declared reliability, durability, and priority. As noted in Section 4.1.1, three schema dimensions (`has_deadline`, `deadline_ns_log`, `max_blocking_ms_log`) remain zero throughout the corpus as reserved extension points; reported gains stem from the four active dimensions alone.

7.3.2. Convergent Validity Over Simulation Oracles

We evaluated inter-oracle agreement across $I^*(v)$ (`FaultInjector`), $I_{\text{comp}}(v)$ (`FailureSimulator`), and $I_{\text{dyn}}(v)$ (`MessageFlowSimulator`) over seven scenarios, summarized in Table 10:

Table 10: Inter-oracle agreement across simulation paradigms (chance top- K Jaccard is 0.111) over 7 diverse benchmark topologies. I_{dyn} denotes the queue-flow discrete-event simulation oracle, I^* denotes the graph topological cascade injection oracle, and I_{comp} denotes the multi-criteria composite oracle. The negative lower bounds on I_{comp} reflect a single scenario (`hub_and_spoke`) driven by tied zero-inflation rather than directional inversion (Supplementary Section S2).

Oracle pair	Mean ρ (range)	Mean τ	Jaccard@ K	Tie-robust
I_{dyn} vs. I^*	0.907 (0.748–0.985)	0.788	0.486	0.509
I_{comp} vs. I^*	0.425 (−0.044–0.654)	0.311	0.240	0.258
I_{comp} vs. I_{dyn}	0.427 (−0.037–0.654)	0.312	0.276	0.275

Ordering converges strongly between the queue-flow simulator and topological cascade oracle ($\rho = 0.907$). Top- K set agreement is more conservative (Jaccard 0.486), bounded by discrete threshold sensitivity in non-linear cascades.

7.3.3. Node-Type Stratification

One result from the detection benchmark governs how every other number in this paper is read. Measured against $I_{\text{comp}}(v)$ over the eight scenarios of that benchmark, stratified RM rank correlations are $\rho = 0.515$ (Application), 0.183 (Broker), and 0.149 (Node), while pooling all types collapses the correlation to $\rho = 0.057$ — below every per-type value it aggregates, which is Simpson’s paradox in its textbook form. This is why every evaluation in this paper is reported on a single stratum, and why pooled critical-set figures should be read as inflated wherever they appear (Section 7.4). The rule-based anti-pattern catalog evaluated on the same benchmark, its behavior under scaling, and the comparison against degree centrality are reported in Supplementary Section S6; its summary is that the catalog flags 94.3% of scored components and therefore does not discriminate, so critical-set identification is delegated to the continuous rankers of Sections 7.1–7.2.

7.3.4. HGT Attention Weight Analysis

Aggregated by relation type over the ATM case study, first-layer mean attention orders **USES** into libraries (0.227) above publish–subscribe channels (0.163–0.176). However, the spread across all eight relation types is narrow (0.15–0.23) and driven substantially by destination in-degree artifacts (such as destinations with in-degree one where $\alpha = 1.0$ holds by definition). Supplementary Section S8 details the full layer-wise attention distribution and heatmap, confirming that multi-head heterogeneous attention remains active across relation types without establishing a statistically distinct relation ordering.

7.4. RQ4: Real-World Distributed Architecture Validation

We evaluated the framework on five open-source distributed systems transcribed from public repositories: Online Boutique, Train-Ticket, Home Assistant, Autoware.universe (ROS 2), and EdgeX Foundry. All carry labels from the same simulation oracles used throughout, testing topological transfer rather than agreement with field failures.

We conduct two distinct real-world evaluations. First, the closed-form explanation layer $Q(v)$ is evaluated against $I_{\text{comp}}(v)$ in Supplementary Section S7, achieving strong correlation across all five systems ($\rho = 0.514$ – 0.800) and outperforming degree centrality. As noted in Section 8.4, **Topo-QoS** is omitted from Table 11 because its projection-level betweenness cache was not computed for the open-source adapters, leaving unweighted **Topo** as the available training-free baseline.

7.4.1. Zero-Shot Transfer of the Learned Model

To test generalization to architectures outside our generator, we trained HGT-QoS on all twelve synthetic scenarios and evaluated it zero-shot across the five open-source systems against $I^*(v)$ (five seeds). To mitigate cross-scenario feature-scale drift, this transfer evaluation applies within-graph rank normalization, 2 message-passing layers, and 150 epochs. No real-world system contributed training gradients or was used for checkpoint selection. Table 11 presents the resulting transfer performance.

Table 11: Zero-shot transfer to five open-source systems, scored against $I^*(v)$ on the Application population. HGT-QoS trains on all twelve synthetic scenarios ($\pm$ = spread over five seeds); RM and Topo are training-free, scored on identical labels and nodes. $\rho_{>0}$ restricts the correlation to the $n_{>0}$ components that actually propagate a failure and is the column to read for ranking quality; full-population ρ conflates that with separating active from inert. **Topo-QoS** is absent (Section 8.4).

Real-World Architecture	$ V_{\text{app}} $	RM ρ	Topo ρ	HGT-QoS ρ	HGT-QoS $\rho_{>0}$	$n_{>0}$	$F_1@K$
Cloud Microservices Mesh	22	0.777	0.891	0.492 ± 0.199	−0.314	18	0.200
Train-Ticket Booking Mesh	41	0.713	0.528	0.702 ± 0.071	−0.244	22	0.375
Autoware.universe (ROS 2)	32	0.357	0.307	0.715 ± 0.067	+0.549	28	0.567
EdgeX Foundry (Industrial IoT)	22	0.470	0.534	0.688 ± 0.086	+0.247	19	0.500
Home Assistant (Smart Home)	24	0.265	0.297	0.803 ± 0.032	+0.564	23	0.480
Mean	—	0.516	0.511	0.680	+0.160	—	0.424

Key Insights for Real-World Transfer:

- The full-population figure is not a ranking result.** On the whole Application population HGT-QoS reaches $\rho = 0.680$ against 0.511 for Topo and 0.516 for RM, leading on four of five systems. Read alone, that looks like successful zero-shot transfer. It is not, because between 4% (Home Assistant) and 46% (Train-Ticket) of Applications in these architectures carry exactly zero simulated impact, and a correlation computed over a population that is heavily tied at zero rewards separating the inert from the active at least as much as ordering the active correctly.
- Restricted to components that actually propagate failures, transfer is not established.** On the active stratum the mean falls from 0.680 to +0.160, and two of the five systems invert: Cloud Microservices to $\rho_{>0} = -0.314$ ($n = 18$) and Train-Ticket to -0.244 ($n = 22$). Both are the microservice call-tree architectures. The three pub-sub systems hold up (+0.549 Autoware, +0.564 Home Assistant, +0.247 EdgeX), so the pattern is architectural rather than random — but a method that anti-correlates with the truth on two of five held-out systems has not demonstrated zero-shot generalization, and we

do not claim it. **We therefore report RQ4 as a negative result:** learned relational transfer to authentic open-source architectures is not established by this evidence.

3. **What the full-population number does support.** Separating components that propagate failures from those that do not is itself the operationally useful half of the task — a gate that correctly identifies which two thirds of a system cannot cause a cascade has narrowed the review surface, even if it orders the remainder poorly. The $F_1@K$ column (0.424 mean) is the honest expression of that capability, and it is what a practitioner would act on. We separate the two claims rather than let the first stand in for the second.
4. **Why the microservice architectures invert.** Both are deep synchronous RPC call trees rather than pub-sub meshes: a large fraction of services are terminal sinks whose failure reaches nobody, and among the minority that do propagate, impact is governed by position in a call hierarchy that the model, trained entirely on generated pub-sub topologies, has never seen. Topo captures the upstream bottleneck structure directly and leads on Cloud Microservices ($\rho = 0.891$). This is a domain-shift limit of the training corpus, not a defect of typed message passing, and it marks the boundary of what our synthetic corpus can prepare a model for.

7.5. RQ5: Analysis Cost and Its Comparison Against Simulation

RQ5 quantifies computational overhead and sustainability during CI/CD evaluation, with per-stage latencies reported in Table 12:

Table 12: Per-stage latency of the inference pipeline across scaling graph sizes (CPU, median of 3 runs).

$ V $	$ E $	Analyze (s)	Graph→tensor (s)	HGT forward (ms)	Analyze : forward
249	1,127	1.74	0.010	26.5	66×
499	2,402	8.32	0.022	16.4	509×
999	6,422	44.54	0.056	21.1	2,108×
1,998	19,301	239.34	0.157	56.2	4,259×

The neural model is the cheapest stage, and the deterministic one is expensive. At 2,000 components the HGT forward pass takes 56 ms against 239 s for deterministic structural analysis — a ratio of 4,259×. That ratio locates cost *inside* the pipeline; it is not the cost of evaluating an architecture. Indices 0–17 of every node feature vector (Section 3.4) — betweenness, closeness, reverse PageRank, articulation and bridge scores — are products of that same analysis stage, so the forward pass cannot run without it. End-to-end evaluation of an unseen 2,000-component architecture costs about four minutes, of which the learned model is 0.02%; the 56 ms figure is the marginal cost of re-scoring an already-analyzed graph. Across the eight-scenario detection benchmark the complete gate (structural analysis plus 18 anti-pattern detectors) runs in 0.04–82.7 s, the upper bound being the 520-component Enterprise mesh.

Cost is dominated by one metric, and it grew. Measured cost now tracks the stage’s $O(|V|^2 + |V||E|)$ bound closely: from 249 to 1,998 components wall-clock rises 138× against a 137× growth in $|V||E|$. The dominant term is the Connectivity Degradation Index, which is computed for every node in the main connected component rather than for articulation points alone. That choice is deliberate and is a correctness requirement rather than an oversight: gating CDI to articulation points leaves it identically zero for every node whose removal does not literally disconnect the graph, which drives $A(v)$ to a near-constant in the redundant multi-publisher topologies this framework targets. The cost is the price of a non-degenerate Availability score, and we report it rather than the cheaper gated variant we could have measured.

7.5.1. The Gate Is Not Cheaper Than the Simulation It Replaces

The framing that motivated this analysis — static gating as a low-cost substitute for dynamic simulation — does not survive measurement against our own oracle. Timing the **FaultInjector** labeling sweep (five seeds, node types Application/Broker/Library, the full ground-truth run) on the same corpus and the same idle hardware gives 0.14–7.2 s per scenario, against 0.04–82.7 s for the analysis gate. Both maxima belong to the 520-component Enterprise mesh, so the largest scenario compares 7.2 s of simulation against 82.7 s

of static analysis: **the gate costs roughly eleven times more than the simulation it is meant to displace.**

This finding refutes the assumption that static analysis is computationally cheaper than in-process simulation: breadth-first cascade traversal is simpler than computing all-pairs connectivity degradation. However, the operational distinction is what the two require: dynamic simulation and chaos engineering demand provisioned clusters and runtime execution, whereas SaG runs directly on manifests during CI/CD. The measurements support the claim that analyzing hundreds of components fits within pull-request budgets without runtime infrastructure, but they do not support general computational superiority over simulation.

8. Discussion, Threats to Validity, and Limitations

8.1. Discussion and Practical Implications

When to use Topo-QoS, and when to use HGT-QoS. The results do not support a simple recommendation of the learned model over the closed-form one, and we set out the trade-off as measured rather than as hoped.

1. **Training-free ranking (Topo-QoS).** It requires no training, no checkpoint storage, and no retraining as a corpus evolves, reaching $\rho = 0.568$ zero-shot across twelve unseen synthetic architectures and ≈ 0.51 across five open-source systems. Nothing in this study establishes that a learned model beats it on ranking: the margin is $+0.127$ ($p = 0.077$) and collapses to $+0.078$ when one fold is removed (Section 7.1). For a team that wants a criticality ordering and nothing more, this is the defensible default, and we say so despite proposing the alternative.
2. **Learned relational prediction (HGT-QoS).** Its established advantage is over *untyped* learning under distribution shift ($+0.114$, 11 of 12 folds), not over closed-form centrality. It also leads on critical-set identification, where it does separate from the training-free baseline ($F_1@K +0.154$, $p = 0.034$). Two further capabilities have no closed-form counterpart, and they are the substantive reason to prefer it where the extra machinery is affordable:
 - *Typed relational attention* exposes *which* channels mediate a cascade, rather than only *which* components rank highly. Supplementary Section S8 illustrates this on one topology and is explicitly not evidence of a general effect.
 - *Relationship-level criticality* (I_{edge} , Eq. (13)) scores individual dependencies rather than components, which is what circuit-breaker or bulkhead placement actually requires, and which a node ranking cannot express. We report this as a property of the formulation rather than an evaluated result: this paper defines the edge oracle and the model’s edge head but presents no evaluation of edge-level predictions against it, so the capability is available and untested.
 - *QoS-conditioned ranking.* The 16-D edge encoding improves out-of-distribution ranking by $+0.054$ within the typed architecture (11 of 12 folds, $p = 0.0093$) and $+0.088$ within the untyped one, so declared middleware contracts carry signal a purely structural score discards.

A gate we can no longer recommend. An earlier version of this work proposed a tiered gate: run the learned model by default and fall back to Topo-QoS when the model’s prediction dispersion $\hat{\sigma}$ fell below a threshold. That heuristic does not replicate: across twelve folds, $\hat{\sigma}$ correlates with the margin over Topo-QoS at $\rho_s = -0.126$ for HGT-QoS (the wrong sign, Section 7.2.1). We therefore withdraw the automated fallback recommendation. Instead, because both engines run in seconds, our results support running both concurrently and escalating ranking disagreements to human architectural review.

Role of the Explanation Layer. The RM attribution profile ($Q(v)$, Section 5) provides transparent, standards-compliant architectural diagnostics aligned with ISO/IEC 25010. By separating single-point-of-failure exposure (Availability) from wide error propagation reach (Fault Tolerance), RM provides qualitative remediation guidance (e.g., distinguishing whether a component requires replication or decoupling) that purely numeric rankers and simulation oracles cannot provide.

8.2. Performance and Computational Sustainability Implications

What sustainability means for a pre-deployment gate. Green software engineering distinguishes the energy a system consumes from the energy its *development and assurance* consume [25], and the machine-learning literature has concentrated on the second: the cost of training and serving models [87, 88, 89], the frameworks that mediate it [90], and how to account for it at all [26, 27]. Reliability assurance sits in the same category and is rarely measured: chaos engineering, hardware-in-the-loop benches, and staging-cluster fault injection consume cluster-hours per sweep, and they do so on every pull request that triggers them. Two findings in this study bear on that, and we state them at the strength the measurements support.

The first is about the learned component, and it is unambiguous. *Within* the pipeline the neural model is negligible: the HGT forward pass is 56 ms on a 2,000-component system against 239 s for deterministic structural feature extraction, a ratio of $4,259\times$ (Section 7.5). Whatever pre-deployment dependability analysis costs, adding a graph neural network is not what makes it expensive. For a special issue asking whether AI techniques can be afforded in a sustainability-conscious engineering process, that is the directly relevant result: here, the AI is the cheap part by three orders of magnitude.

The second is about what the framework removes, and it is a difference in kind rather than in degree. The resource a manifest-time analysis eliminates is *infrastructure*, not CPU seconds. Chaos engineering and fault injection on a staging cluster require a provisioned, running system; SaG requires a file. Nothing must be deployed, kept warm, or torn down, and no fault is injected into anything a user could be holding. That is the honest form of the sustainability argument — but we have not measured a chaos-engineering baseline, so its magnitude is an assertion rather than a result.

The efficiency claim we withdraw. As quantified in Section 7.5.1, static analysis does not reduce computation relative to in-process simulation: global connectivity degradation (82.7 s on the Enterprise mesh) is roughly eleven times slower than BFS cascade traversal (7.2 s). We explicitly withdraw any general claim of computational efficiency relative to simulation; the operational saving rests entirely on eliminating the need for live, provisioned staging infrastructure.

Where the cost actually sits, and why we paid it. The expense is concentrated in one metric and is a deliberate accuracy purchase. The Connectivity Degradation Index dominates because it is computed for every node in the main component rather than for articulation points alone; gating it to articulation points would restore roughly an order of magnitude of speed at the price of a degenerate Availability score (Section 7.5). A deployment that needed the speed more than the single-point-of-failure sensitivity could make the opposite trade, and we would rather document that choice than hide it behind a faster number. This also identifies where an optimization effort should go: the sustainability of this framework is a question about one graph-theoretic routine, not about its use of machine learning.

Settling this comprehensively requires energy counters rather than wall-clock time [26]. Furthermore, while preventing cascading failures plausibly reduces datacenter compute consumed by retry storms and restart loops, we treat this as motivation rather than an empirical claim.

8.3. Threats to Validity

Construct Validity. Our primary ground-truth impact oracle $I^*(v)$ is derived from discrete-event cascade simulation on structural models rather than live outages. Evaluating construct divergence against the queue-flow discrete-event simulator $I_{\text{dyn}}(v)$ and composite oracle $I_{\text{comp}}(v)$ (Table 10) reveals high correlation with I_{dyn} ($\rho = 0.907$), which traverses identical failure semantics. However, agreement with I_{comp} is moderate ($\rho = 0.425$), and top- K critical-set Jaccard reaches only 0.24–0.49 due to non-linear cascade threshold sensitivity and tied zero-inflation (Supplementary Section S2). Thus, convergent validity across simulation paradigms holds for ranking but is weaker for critical-set boundaries; no oracle is validated against live outages (Section 4.4). For Maintainability, $I_M(v)$ traverses the same dependency topology from which $M(v)$ is scored; independent churn validation remains future work.

Internal Validity. Potential feature leakage is prevented by strict graph view separation: predictors operate exclusively on G_{analysis} , whereas ground-truth simulation oracles operate on $G_{\text{structural}}$, formally asserted in continuous integration. Substrate parity is rigorously maintained: learned models (HGT-QoS, GAT-N-QoS) share identical training sets, depths, and held-out scenario early-stopping protocols. In the QoS schema, four dimensions (reliability, durability, transport priority, heterogeneity flag) capture active operational middleware configurations, while three dimensions (deadline, max blocking) are reserved extension points that remain zero in standard static descriptors.

External Validity. Our evaluation spans twelve synthetic architectures and five open-source systems. Zero-shot transfer to real systems is *not* demonstrated on active components (mean $\rho_{>0} = +0.160$, inverting on the two microservice call trees, Section 7.4.1). Because the training corpus consists of pub-sub meshes, deep synchronous RPC hierarchies represent out-of-domain topologies that additional pub-sub data cannot resolve. Furthermore, while synthetic QoS profiles exhibit genuine variance (modal shares 29–89%, Section 7.3.1), their alignment with production distributions remains unverified. Finally, our timing evaluations scale to 2,000 components; architectures an order of magnitude larger cannot be assumed to fit PR budgets given the $O(|V|^2 + |V||E|)$ dominant stage.

Conclusion Validity. Given heavy-tailed impact distributions, statistical analyses use non-parametric rank correlation (Spearman ρ , Kendall τ), bootstrap confidence intervals ($B = 2,000$), and paired Wilcoxon signed-rank tests, with the fold or scenario as the unit of analysis. Two hazards recur and shape how we report. Pooling across heterogeneous entity types triggers Simpson’s paradox — pooled $\rho = 0.057$ sits below every per-type value it aggregates (0.149–0.515) — so all headline figures are stratified on a single population. And rank correlation over zero-inflated labels conflates ordering the active components with separating them from inert ones, which is why we report zero-excluded correlations alongside full-population ones wherever the label distribution permits. Where the two disagree, as on the real-world systems, we read the zero-excluded figure as the ranking result.

8.4. Limitations and Future Work

An incomplete baseline on real-world systems. **Topo-QoS** is omitted from Table 11 because its projection-graph cache was not preserved. Rather than report unreproducible figures, we compared against **Topo**; extending the cache is an immediate priority.

The explanation layer is not validated as an explanation. SaG separates Availability from Fault Tolerance on the hypothesis of distinct repairs, but we do not evaluate whether practitioners find this actionable. Furthermore, elicited AHP weights perform worse than a uniform prior at ranking (Section 7.3); counterfactual mutation tests and user studies are needed for validation.

Model selection is made on the training distribution. Early stopping in LOSO uses an inner validation split within the primary training graph (Section 6.3). While a protocol limitation under distribution shift, the rule is uniform across variants and cannot manufacture the typed-versus-untyped margin. Held-out scenario validation (`--inner-val-scenario auto`) is our prioritized extension.

No label-free reliability signal. Prediction dispersion does not replicate as a fallback indicator on this corpus (Section 7.2.1); discovering a reliable OOD confidence signal remains an open problem.

Future Directions: Distributed AI, Power Testbeds, and Self-Healing. We envision three primary extensions: (1) modeling distributed LLM serving clusters (e.g., vLLM, DeepSpeed) to mitigate straggler-induced GPU dissipation; (2) measuring hardware energy directly via RAPL/NVML to compare static gating against live chaos sweeps in joules; and (3) advancing from predictive diagnostics to prescriptive synthesis, automatically generating pull requests with circuit breakers, broker replicas, and tuned QoS parameters to resolve single points of failure.

9. Conclusion

This work introduced **Software-as-a-Graph (SaG)**, a pre-deployment Static System Analysis framework that combines a relation-specific Heterogeneous Graph Transformer for failure-impact forecasting with an interpretable ISO/IEC 25010 Reliability–Maintainability attribution layer, both operating on a typed multigraph derived from Architecture-as-Code manifests with no runtime telemetry.

The central empirical finding is about *when* architectural typing pays. Given training data from the same distribution as the test split, typed and untyped message passing are indistinguishable, and the best in-distribution mean belongs to the untyped model. Asked to rank an architecture it has never seen, the typed model wins 11 of 12 inductive folds by $\Delta\rho = +0.114$ ($p = 0.0122$), and the 16-D QoS edge encoding adds a further $+0.054$ ($p = 0.0093$) independently of typing. Relation-specific parameters therefore act as an inductive bias for unfamiliar topologies rather than as additional fitting capacity — which is the regime a pre-deployment gate always operates in, since every architecture it sees is by construction new.

We are equally clear about what is not established. Against an unparameterized QoS-weighted centrality score, learned ranking is not superior: the nominal margin of $+0.127$ ($p = 0.077$) rests almost entirely on a single fold and falls to $+0.078$ without it. On the five open-source systems, full-population correlation of $\rho = 0.680$ drops to $+0.160$ once restricted to components that actually propagate failures, and inverts on both microservice call-tree architectures, so zero-shot transfer to authentic systems is a negative result on this evidence. A label-free confidence signal we previously reported does not replicate. The explanation layer’s elicited AHP weights measurably worsen ranking relative to a uniform prior, and we have no independent evidence that they improve attribution — validating the attribution itself remains its principal open question.

What the work does contribute is a reproducible typed-multigraph formulation of pub-sub architecture, a corpus that regenerates byte-identically from committed configurations, a clean measurement of where relational typing helps and where it does not, and a pre-deployment analysis pipeline whose learned component is the cheapest stage by three orders of magnitude. Whether that pipeline predicts failures that actually occur — rather than failures a simulator produces — is the question we most want answered next, and it requires field data no static corpus can supply.

Declarations

CRedit Authorship Contribution Statement. **Ibrahim Onuralp Yigit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feza Buzluca:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision.

Declaration of Competing Interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Material. Parameter sensitivity analyses (OFAT, Morris screening, threshold and normalization sweeps, zero-inflation diagnostics, and HGT attention distributions) are provided in the online supplementary document (Sections S1–S8).

Data Availability. The complete replication package—including synthetic datasets, generator configurations, simulation harnesses, real-world adapters, model checkpoints, and analysis scripts—is openly available on Zenodo under DOI 10.5281/zenodo.14922108 [91]. Synthetic datasets regenerate byte-identically from committed configurations. All 177 reported table values are deterministically verified against backing JSON artifacts via the included `reproduce/reconcile_manuscript.py` script.

Declaration of Generative AI in the Writing Process. During the preparation of this work, the authors used AI-assisted language tools to check grammar, improve readability, and support LaTeX typesetting. The authors reviewed and edited all content and take full responsibility for the published article.

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